

影像處理 02 數位影像基礎

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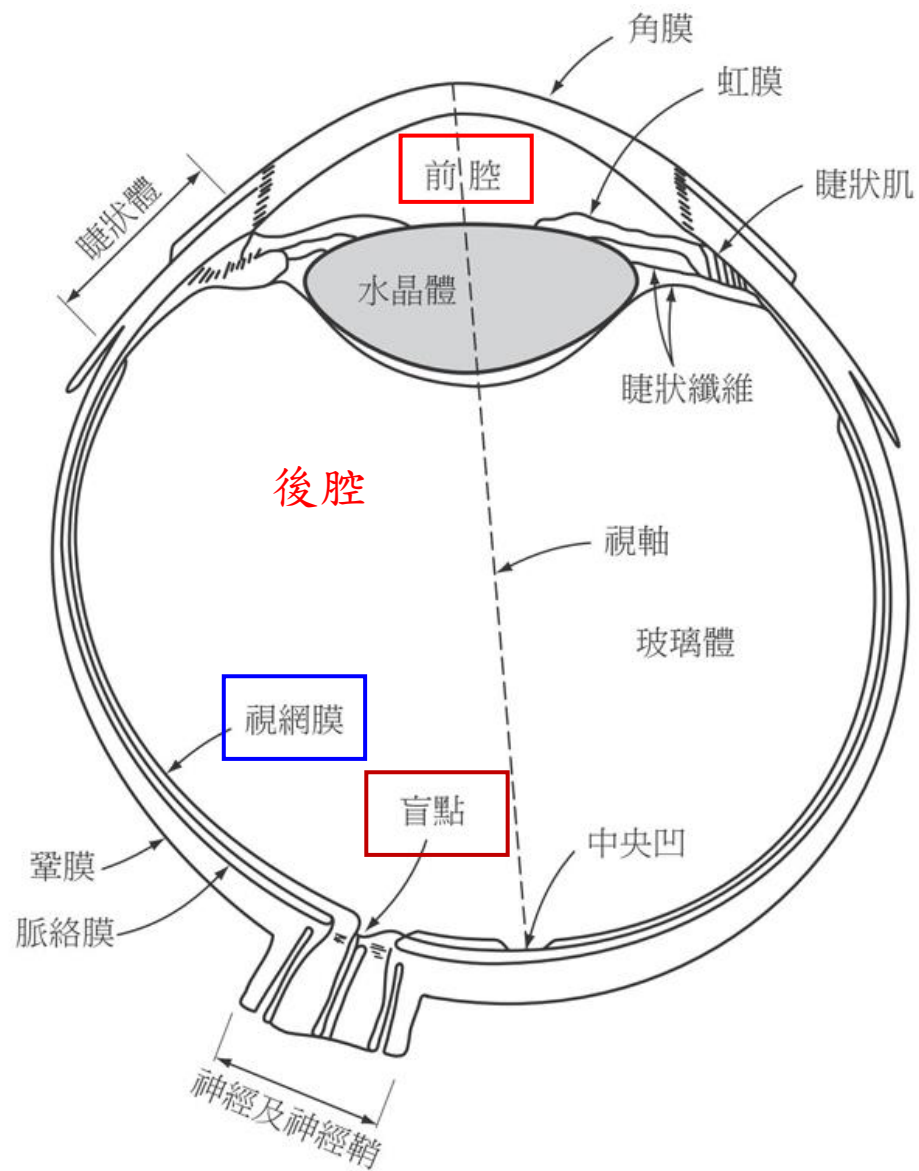
助教:陳信宏

大綱

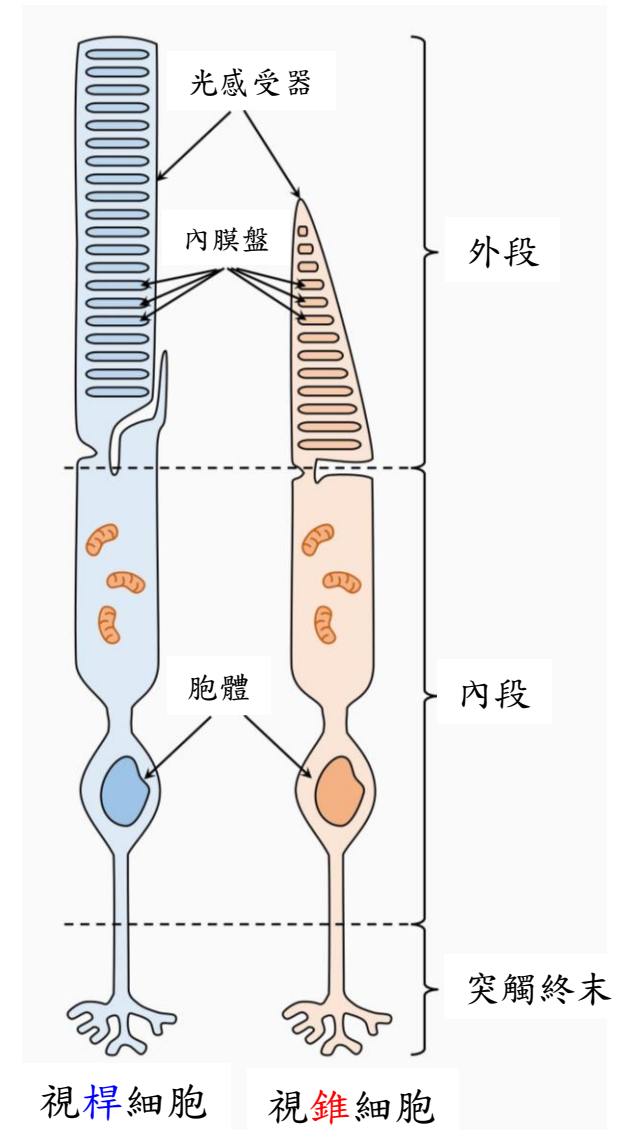
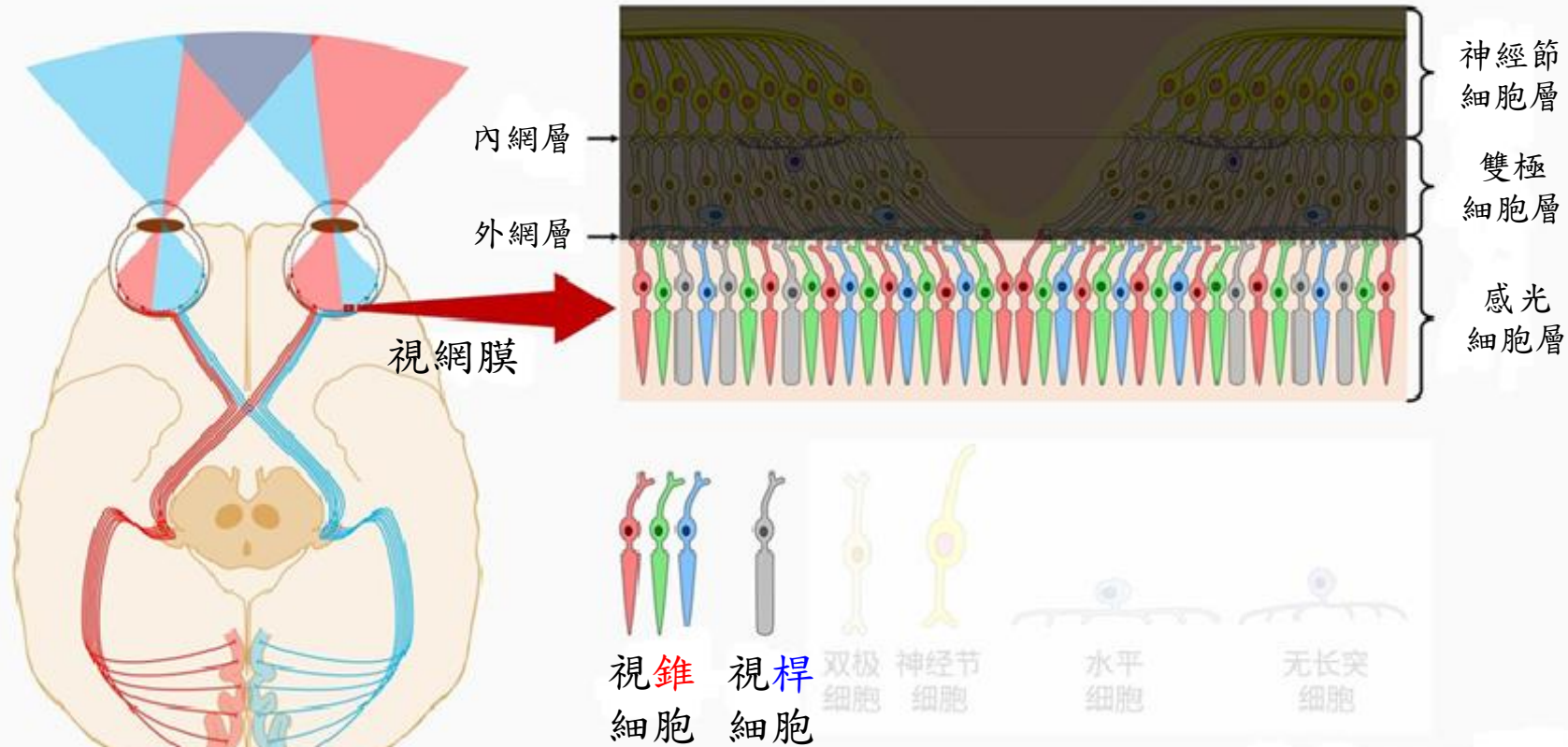
- 視覺感知
- 影像感應與擷取
- 影像取樣與量化
- 內插法(Interpolation)
- 空間轉換

2.1 視覺感知的要素

圖 2.1
人類眼睛橫剖面的
簡化圖形



桿細胞(Rod) & 錐細胞(Cone)



細胞分布

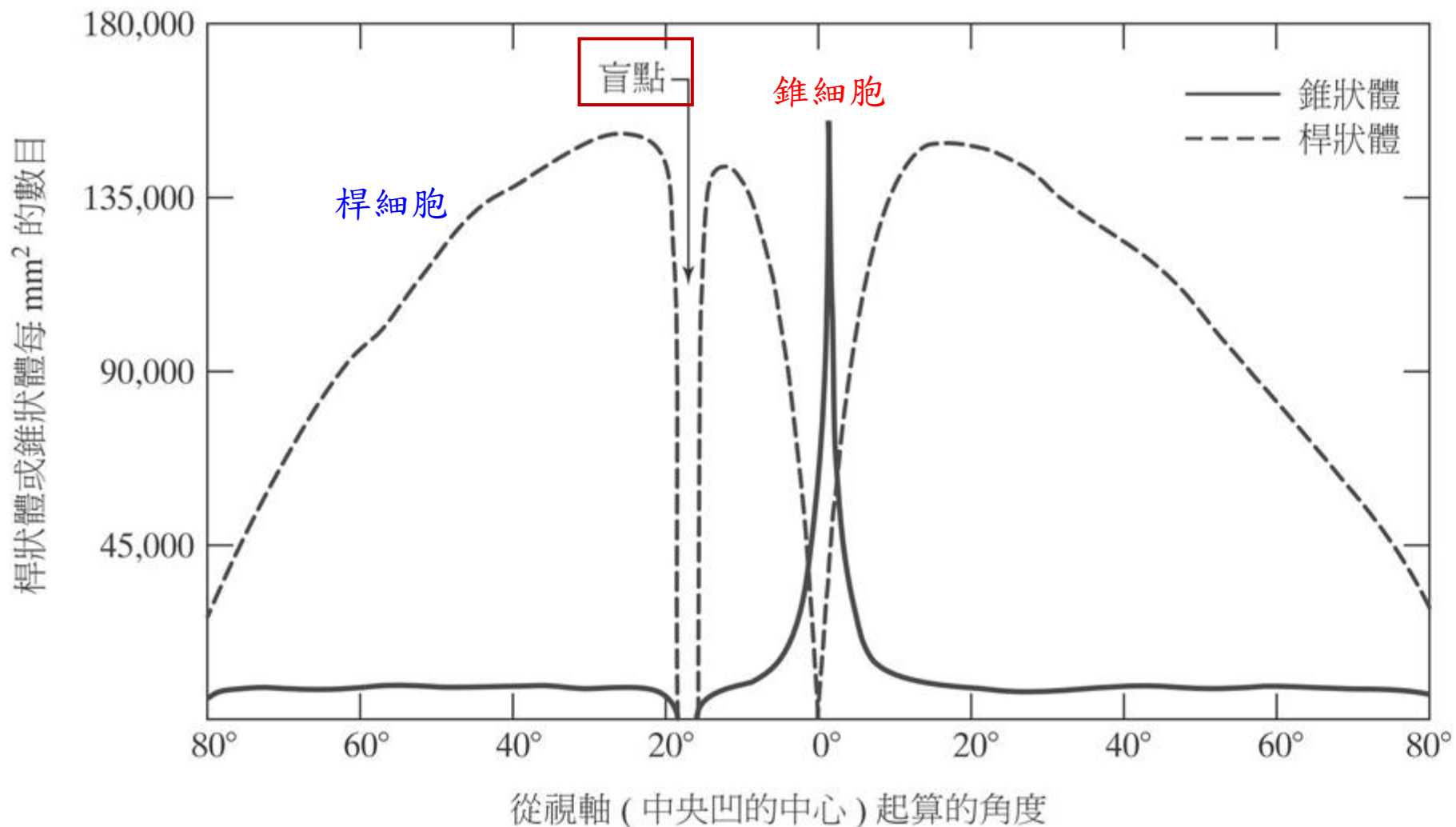


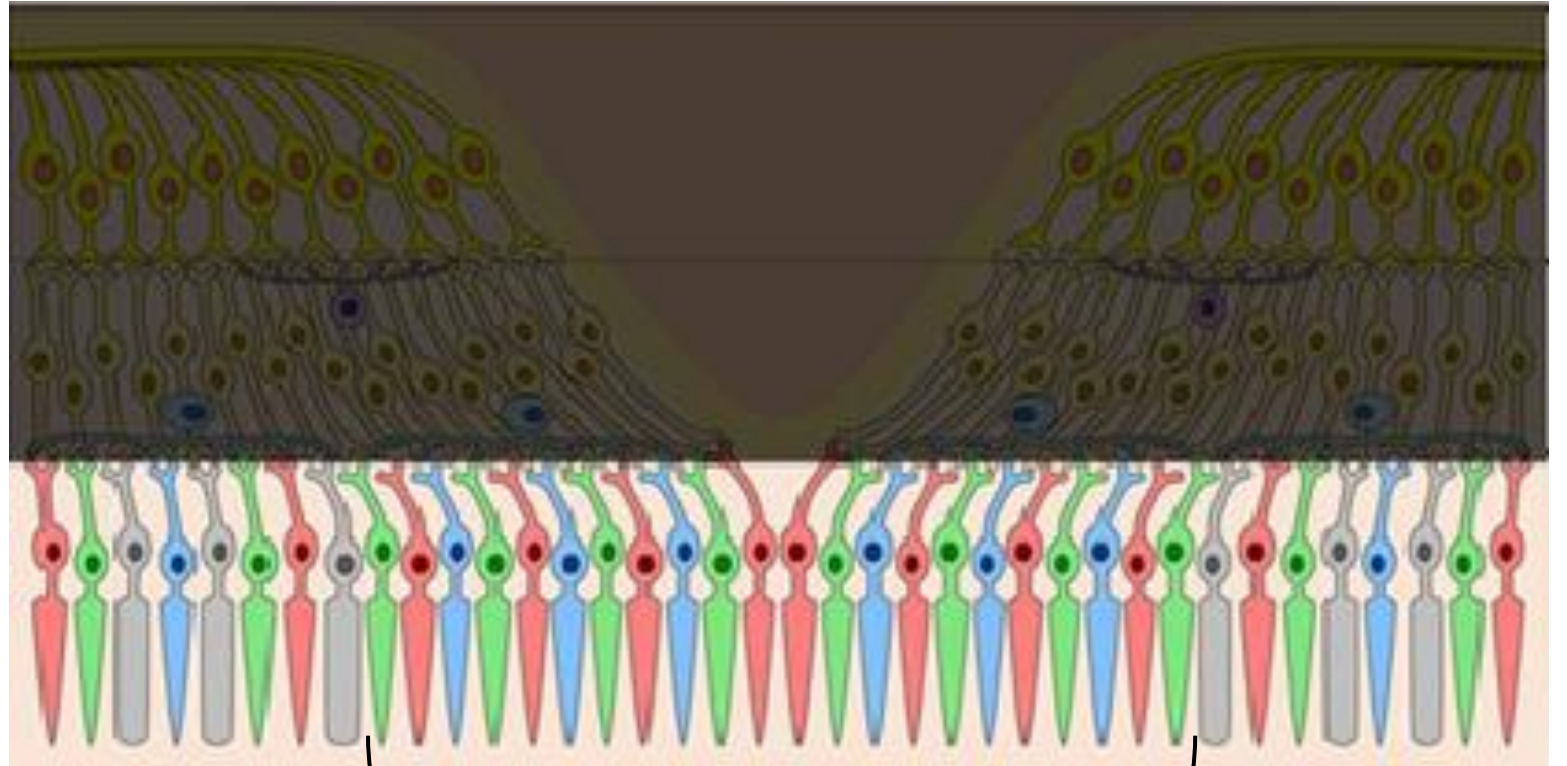
圖 2.2

視網膜上桿狀體
和錐狀體的分布

錐細胞(Cone):
在亮光下工作
負責顏色與細節辨識

桿細胞(Rod):
在昏暗工作
負責光強/區塊辨識

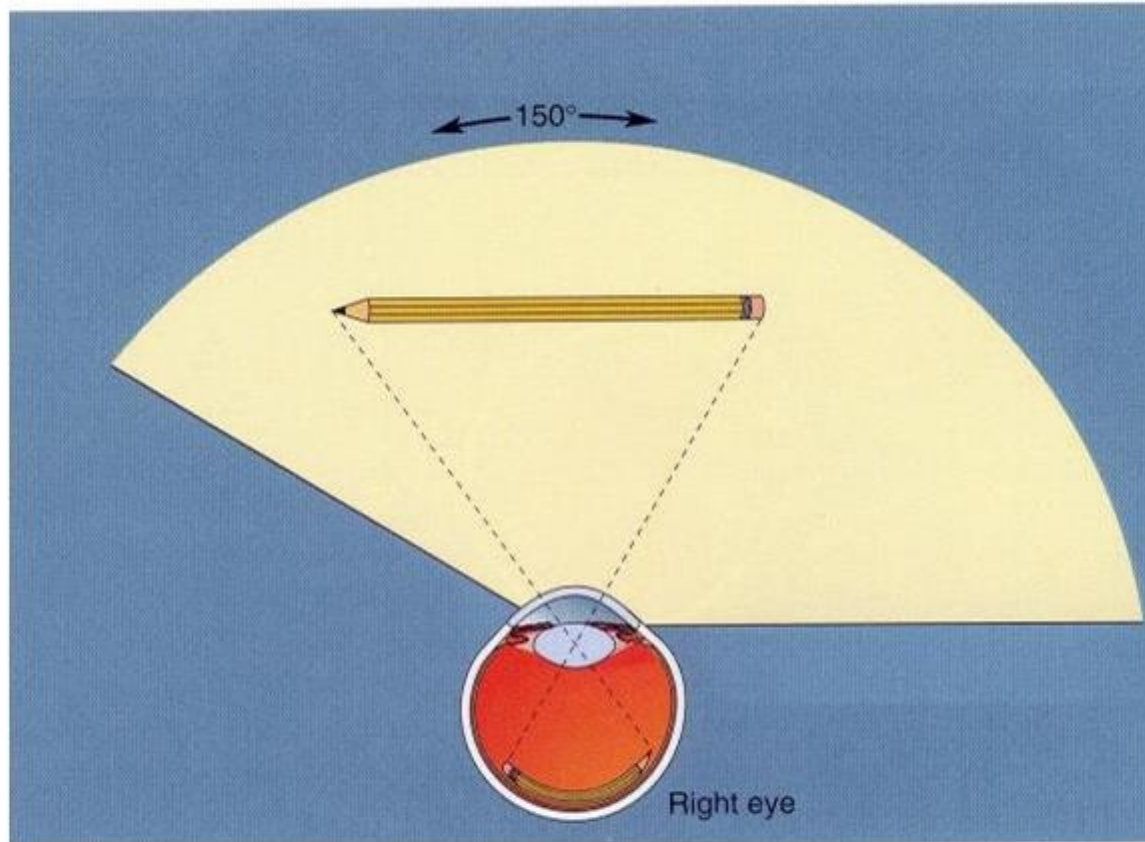
細胞分布



中間都是錐細胞

所以視覺聚焦看細節 都是眼睛要看過去

The visual field

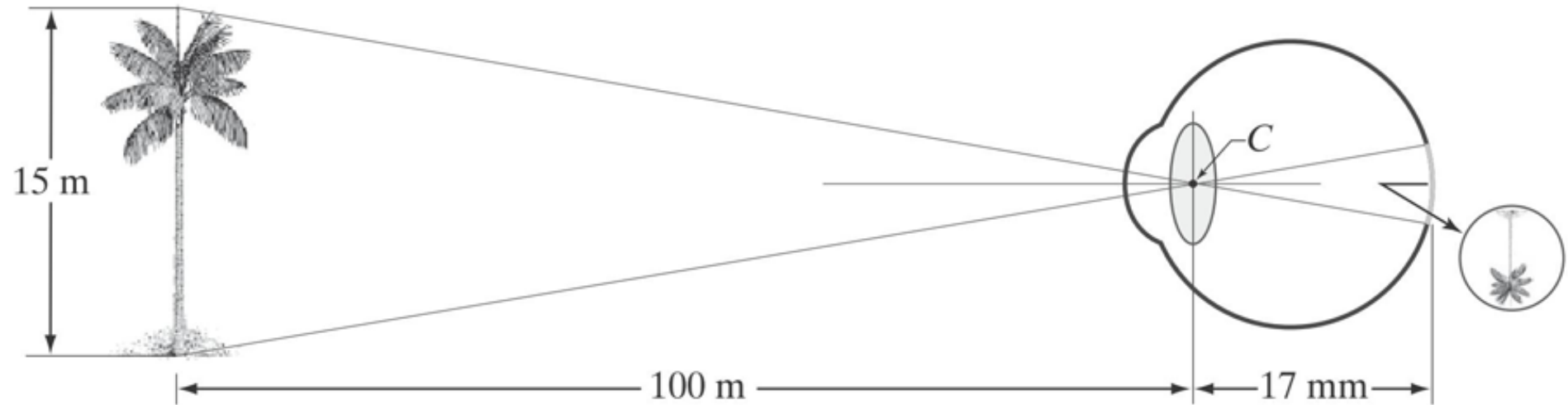


The image is **inverted**; the **left visual field** is imaged on the **right side of the retina**, and the right visual field on left side of the retina

眼睛中的細胞

圖 2.3

注視一棵棕櫚樹時眼睛的圖形表示。 C 點是水晶體的聚焦中心。



$$\frac{15}{100} = \frac{h}{17}$$

$$h = 2.25\text{mm}$$

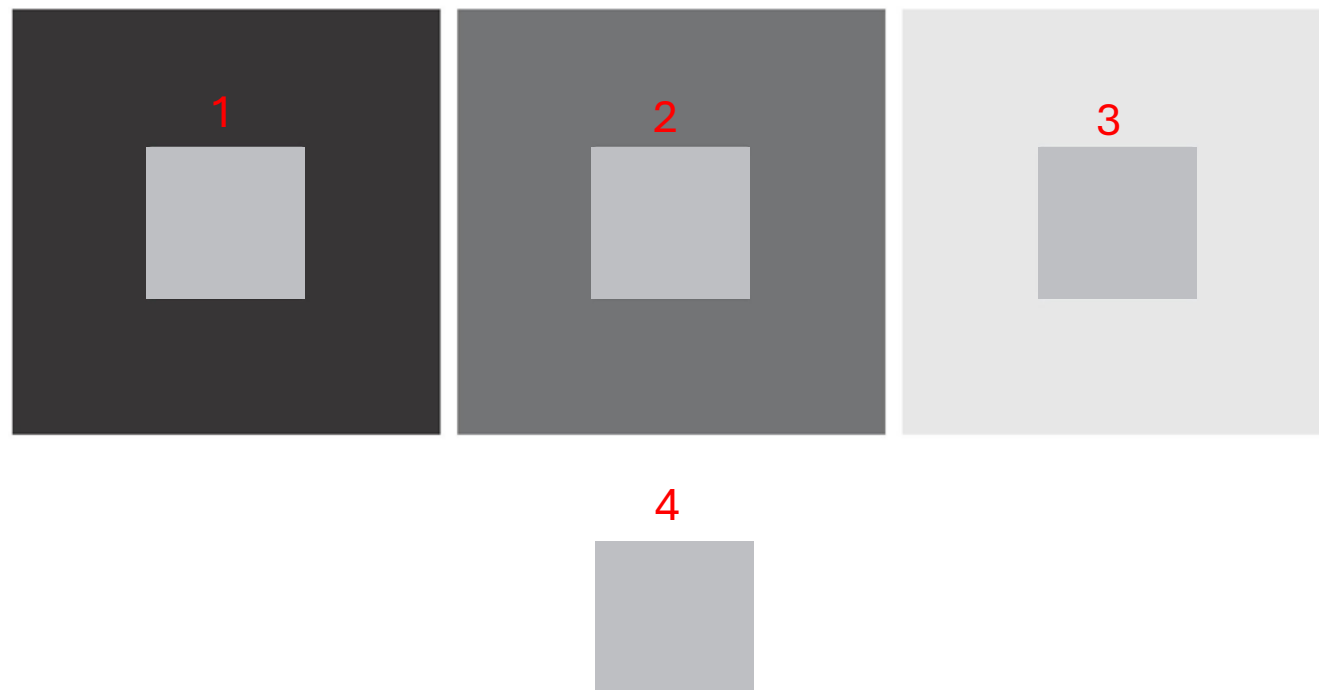
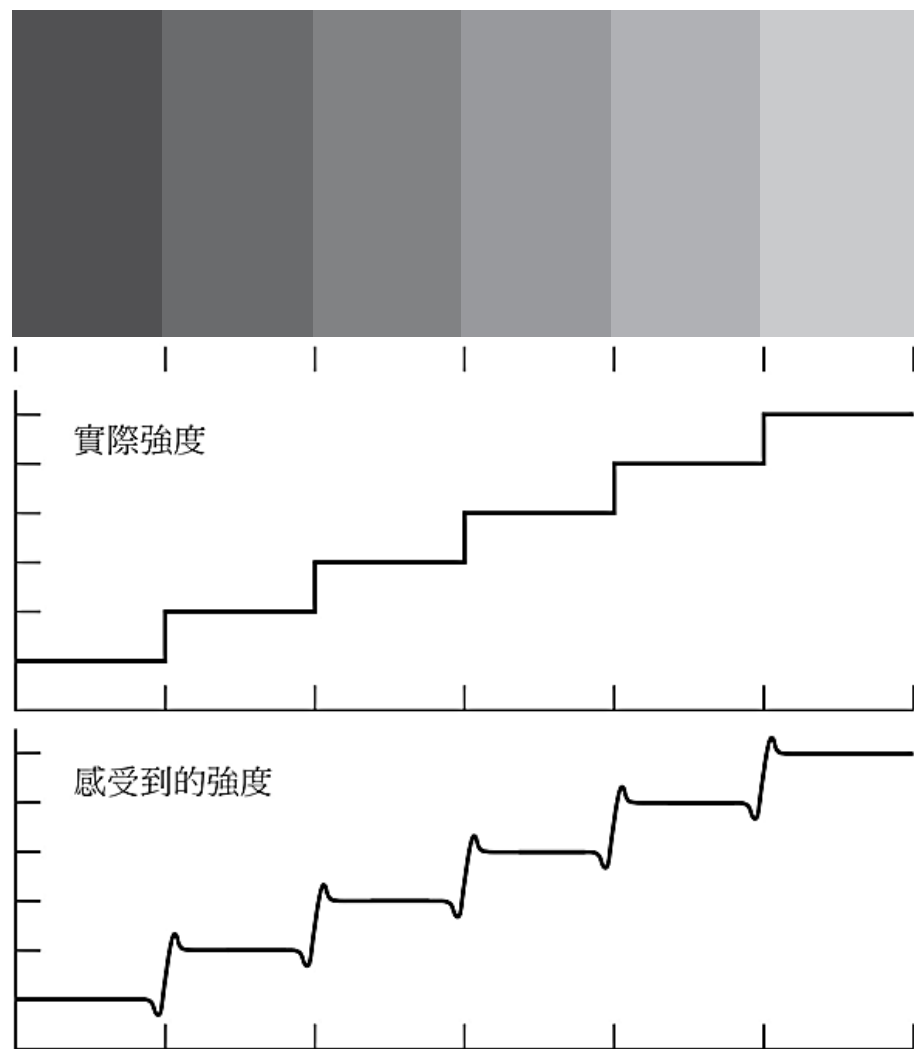
<https://www.youtube.com/watch?v=0HIYbNPCVts>

盲點實驗 | Find Your Visual Blind Spot

可能要近一點看

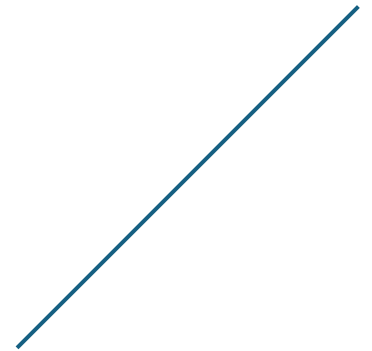
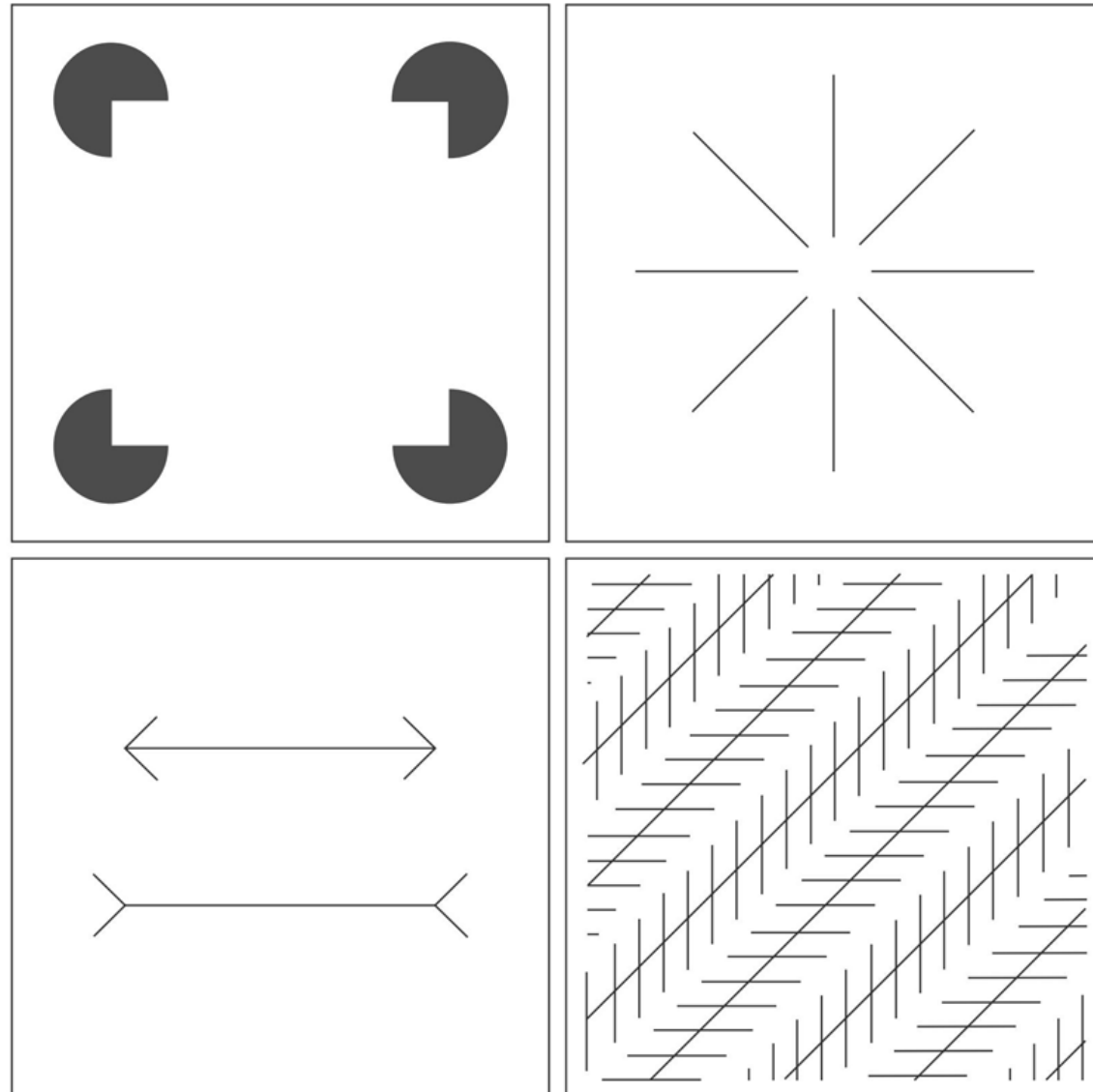


人眼的視覺假象(1/2)



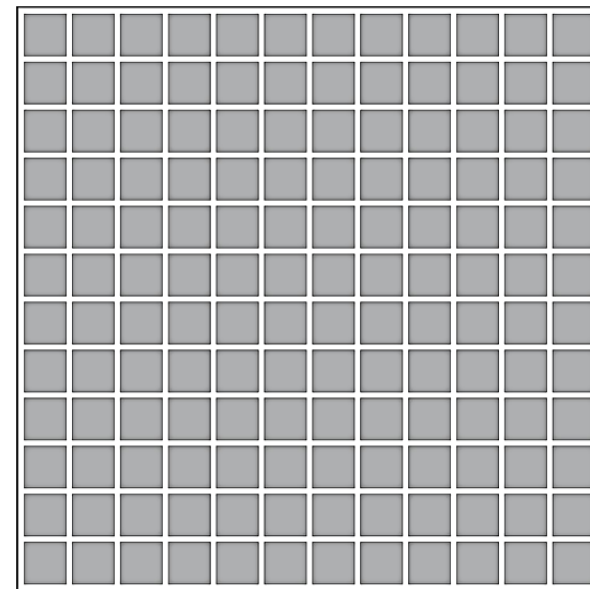
- (A) 1234 都不一樣
- (B) 12 一樣 34 一樣
- (C) 1234 都一樣

人眼的視覺假象(2/2)

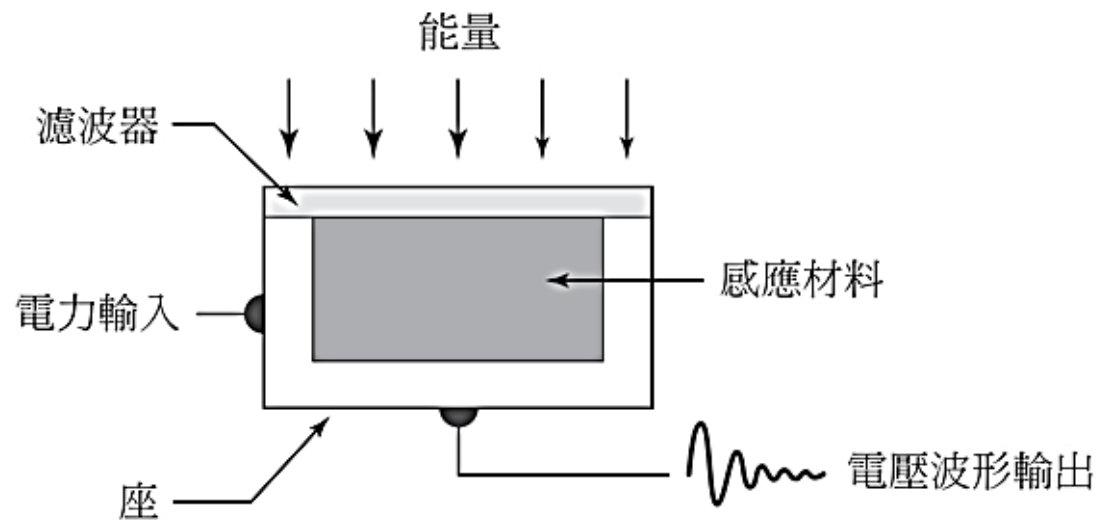


2.2 影像的感應與擷取

面影像感測器

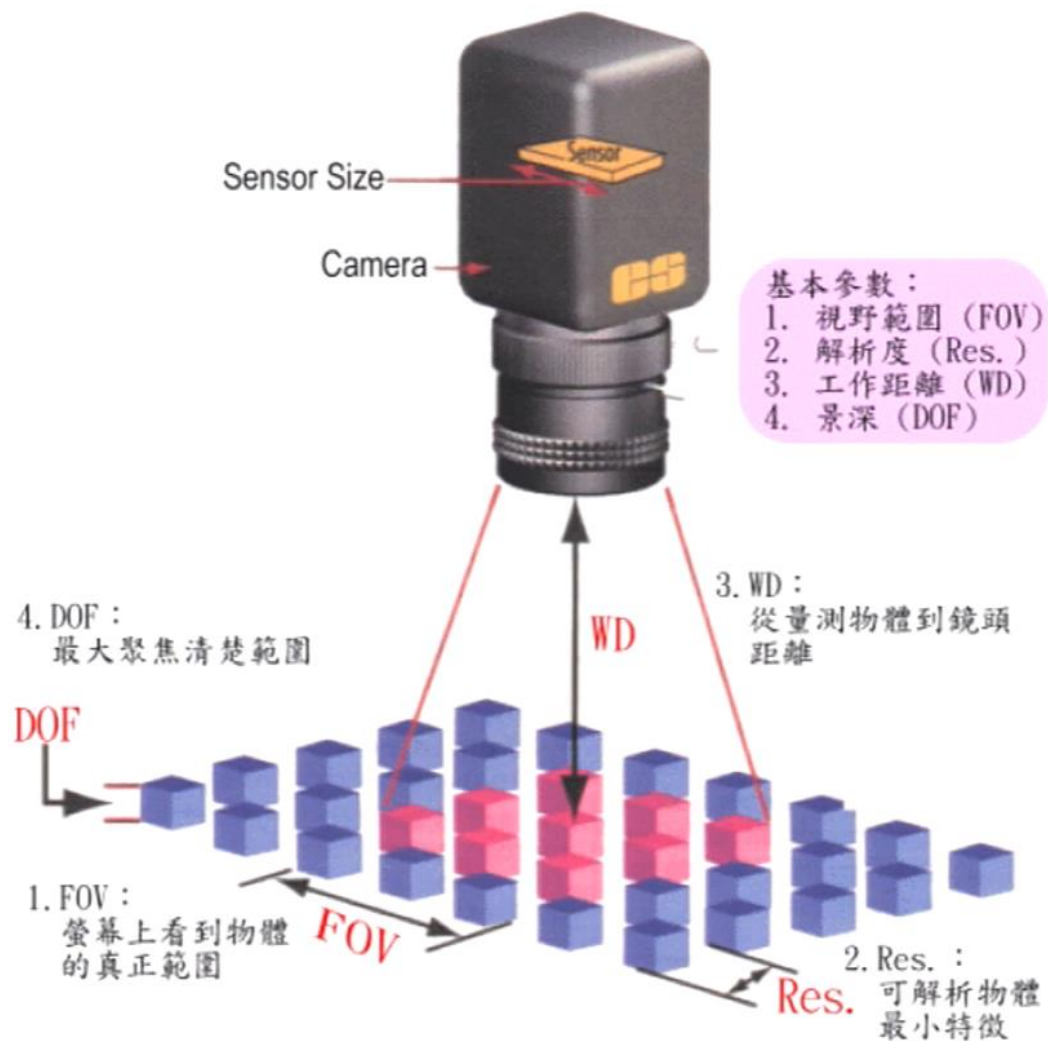


線影像感測器



面影像感測器

➤ 數位相機的 CMOS、CCD

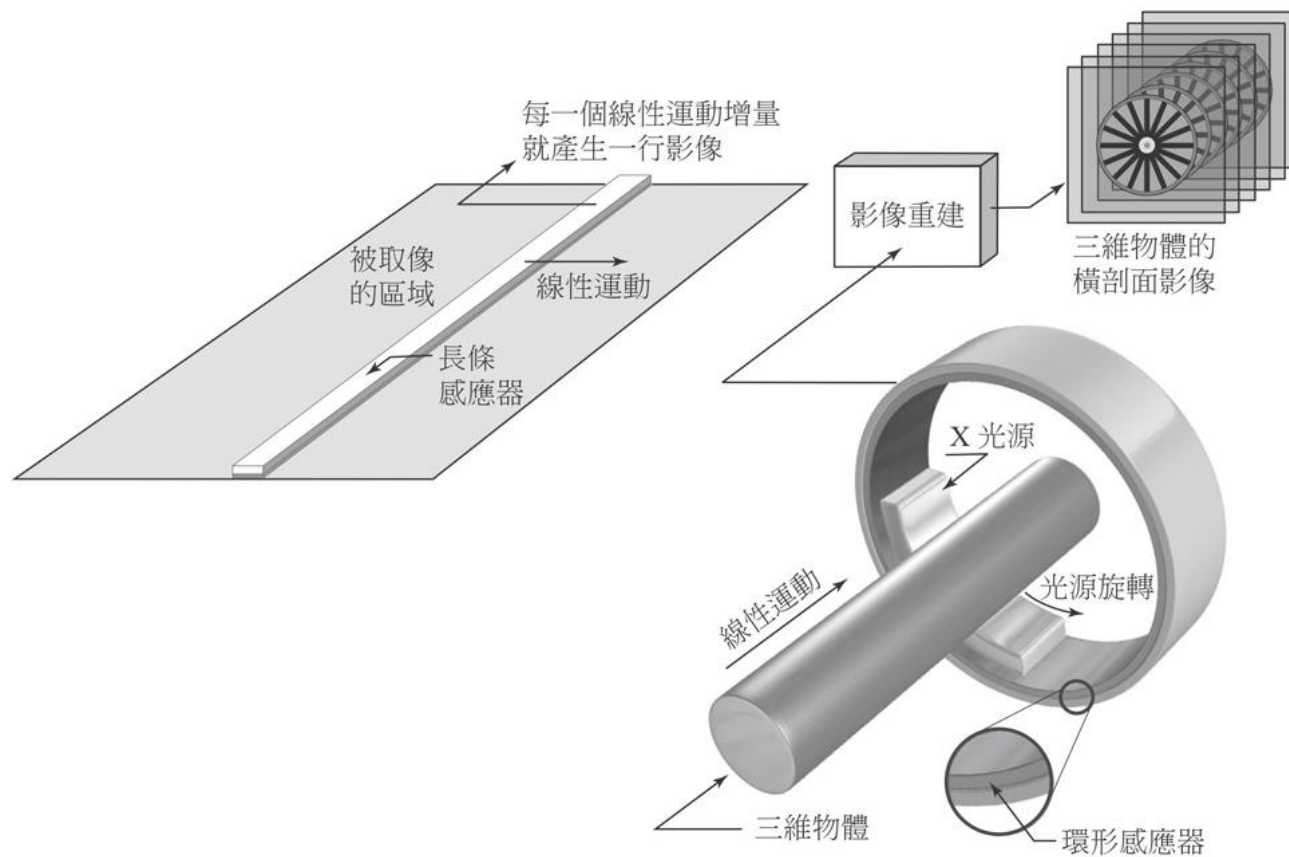


線影像感測器

- 刷載具
- 無人機空中成像
- 核磁共振MRI
- PET



@SOGI.com.tw

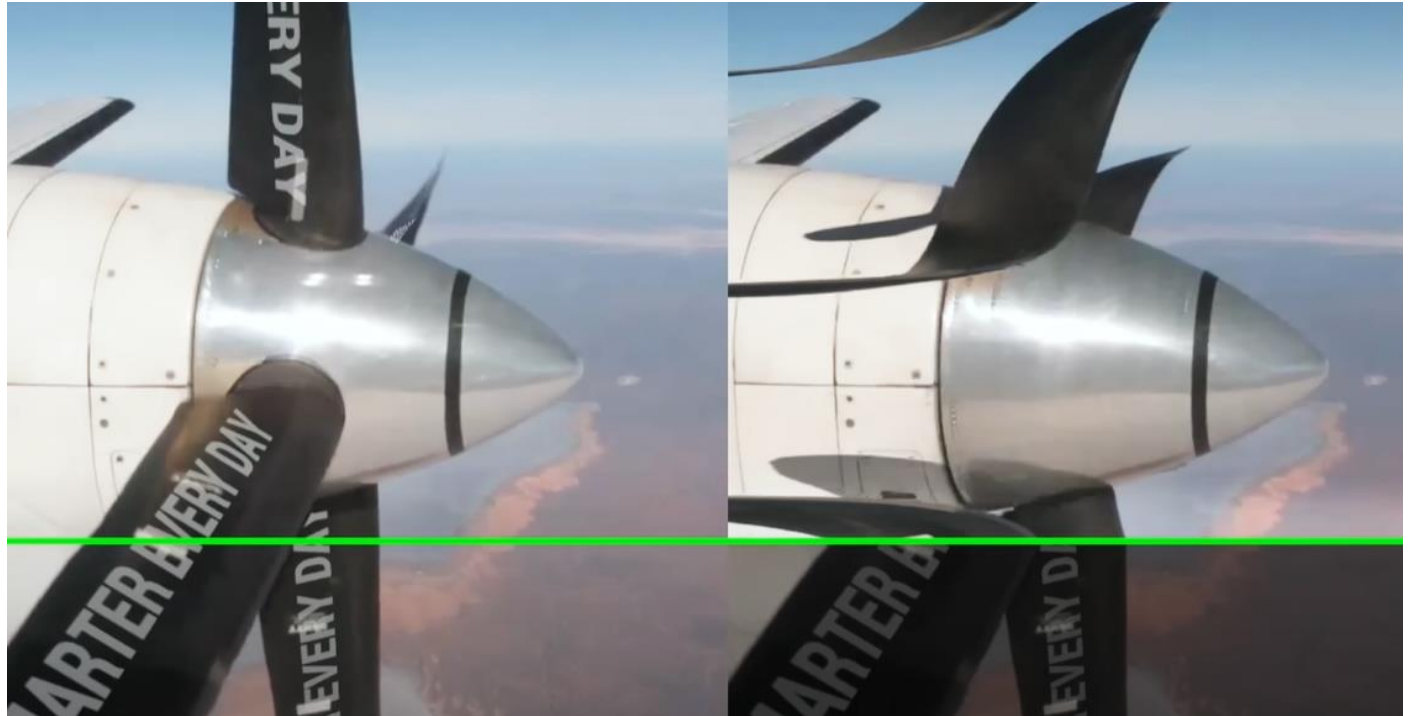


a b
圖 2.12
(a) 用一個長條線型感應器的影像擷取；(b) 用一個圓形長條感應器的影像擷取。

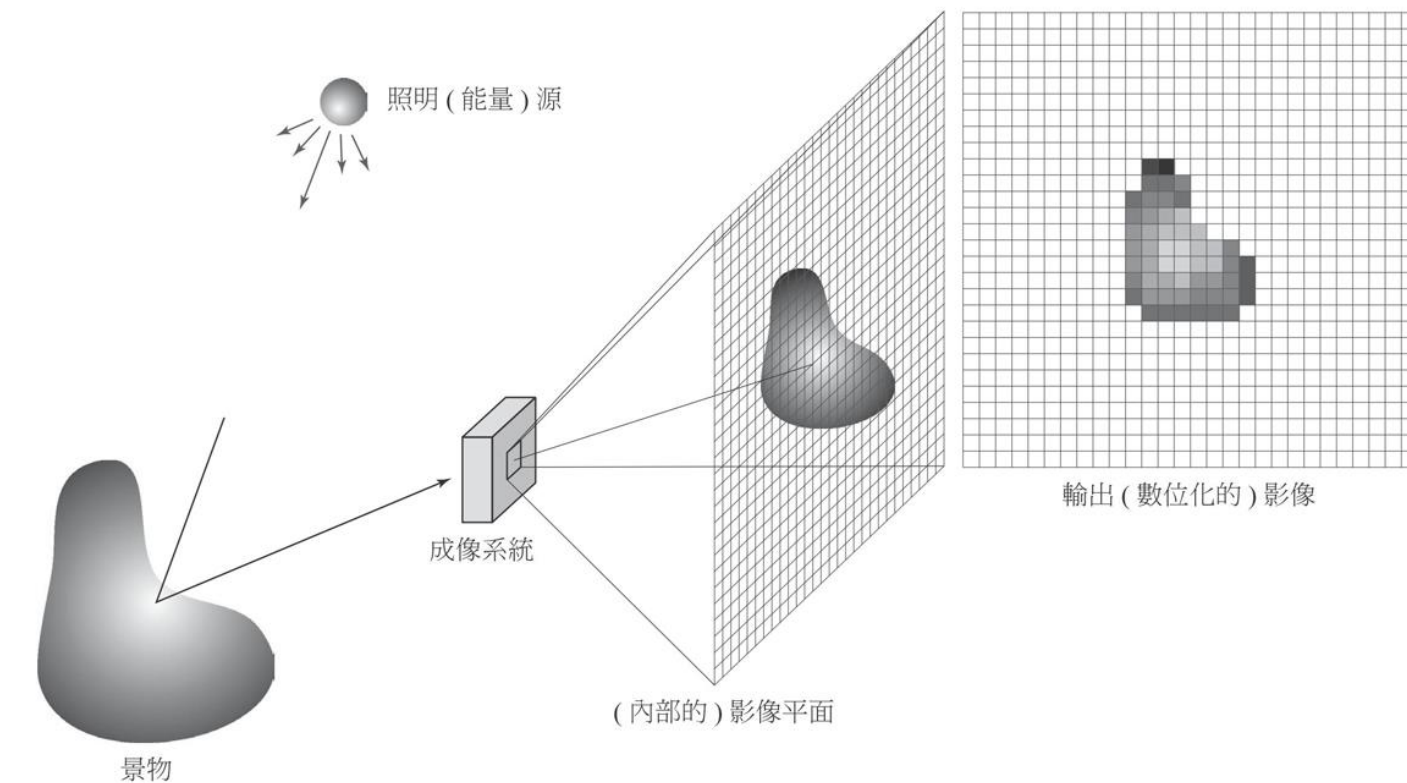
Rolling Shutter Effect

<https://www.youtube.com/watch?v=dNVtMmLlnoE&t=33s>

- 面影像感測器 (Area sensor) 也會有



2D影像形成模型



a
b c d e

圖 2.13 一個數位影像擷取的例子：(a) 照明 (能量) 源；(b) 一個景物；(c) 成像系統；(d) 將景物投影到影像平面上；(e) 數位化的影像。

“理論上”

$$0 \leq f(x, y) < \infty$$

亮度值 (intensity) = 照明量 × 反射量

$$f(x, y) = i(x, y)r(x, y)$$

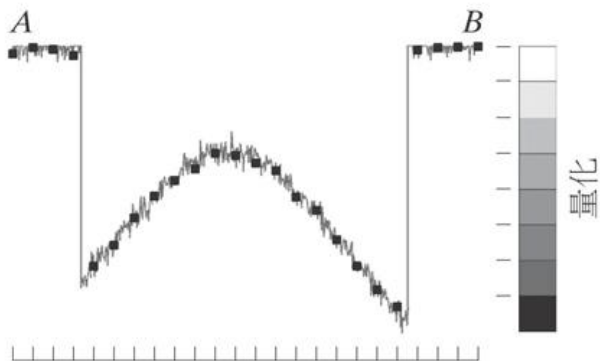
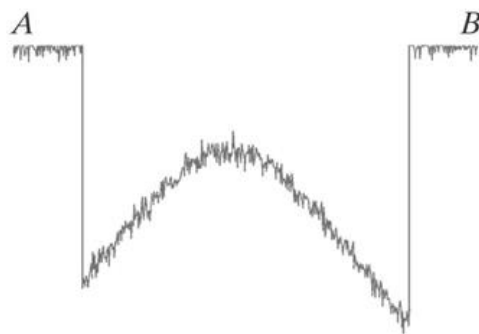
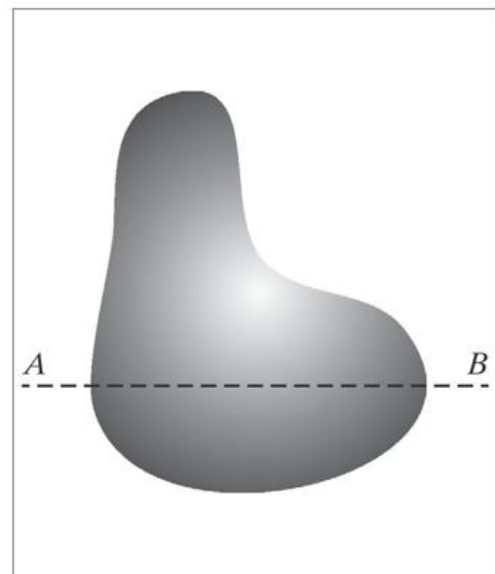
illumination

$$0 \leq i(x, y) < \infty$$

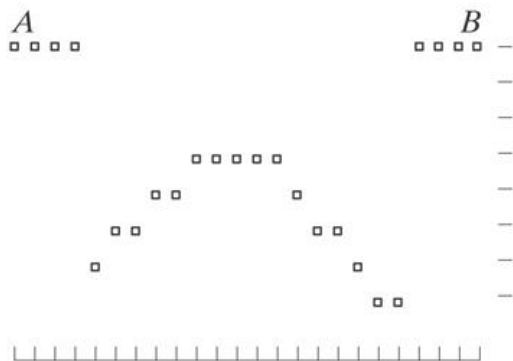
reflectance

$$0 \leq r(x, y) \leq 1$$

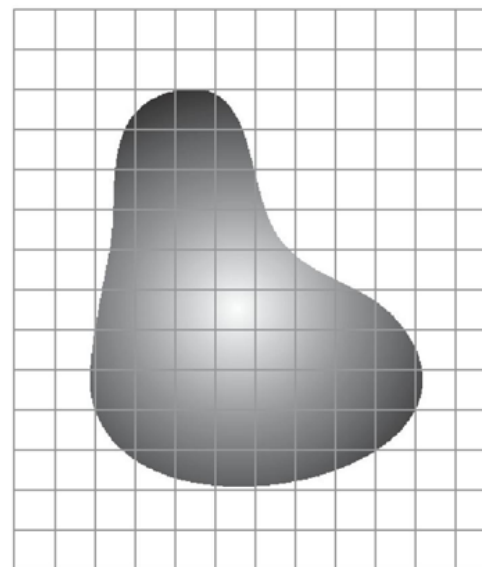
2.3 影像的取樣和量化



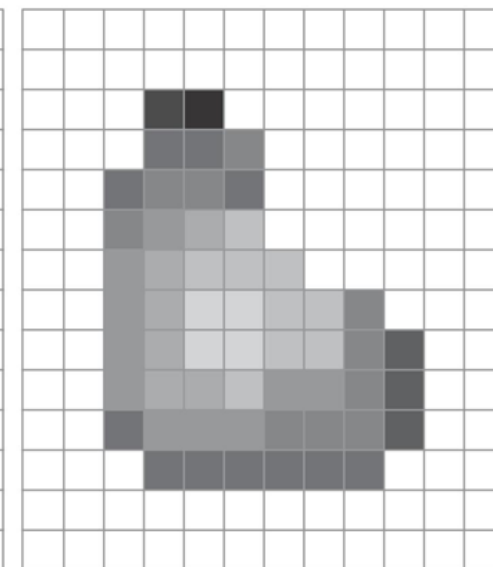
取樣



量化前



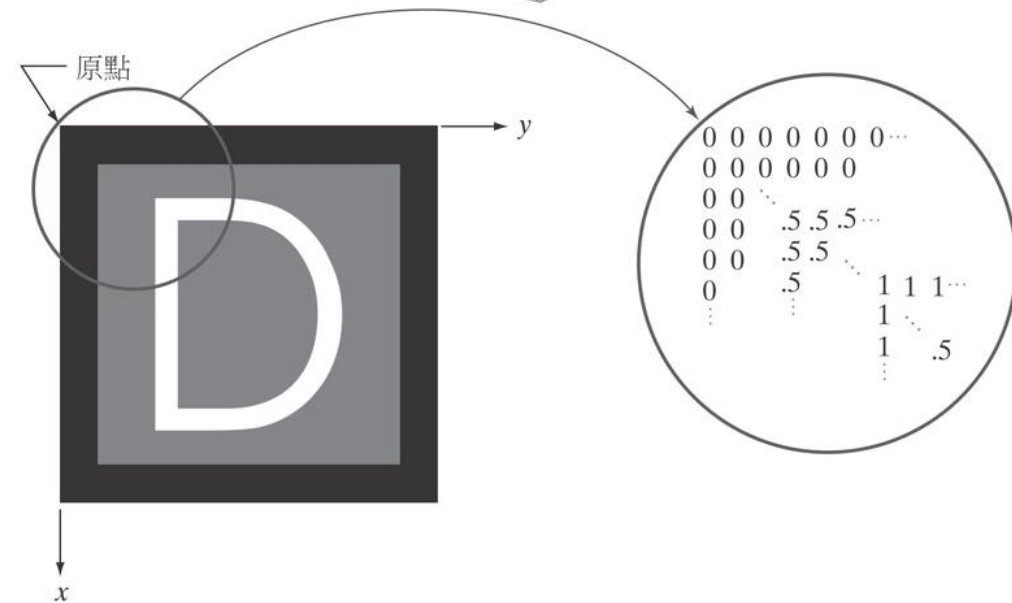
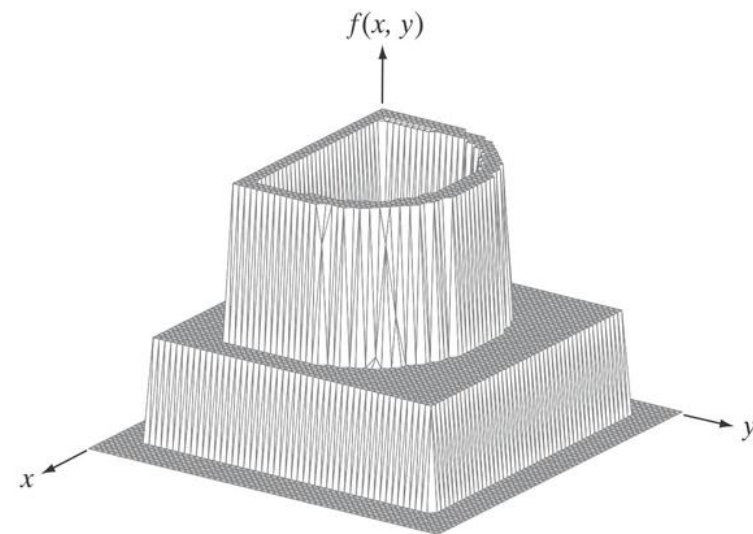
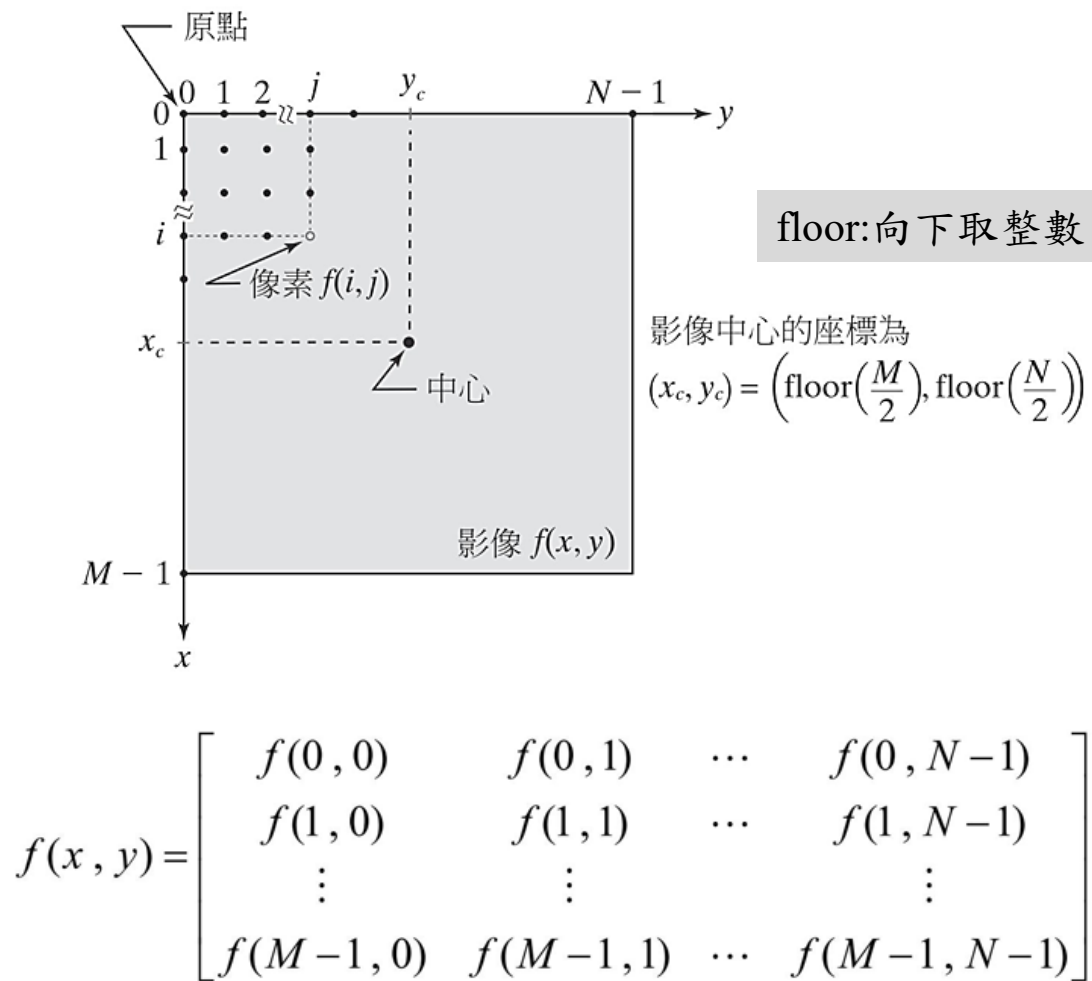
量化後



(8-bit 影像): $0 \leq f(x, y) \leq 255$



數位影像表達

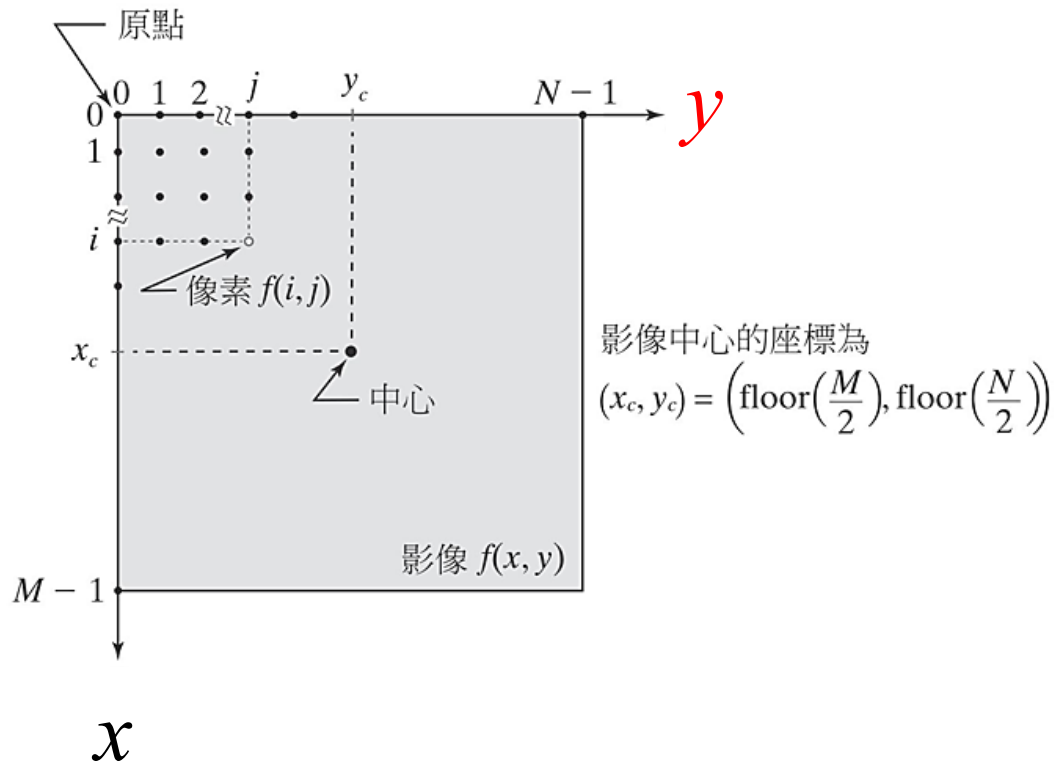


CRT映像管(陰極射線) 原理

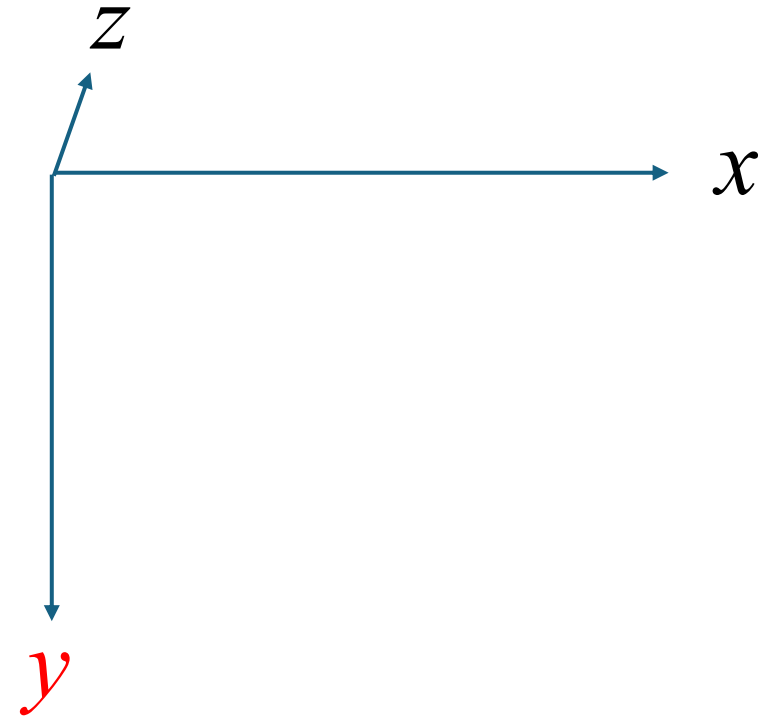
<https://www.youtube.com/watch?v=8ykNPRkmmmbQ>

影像座標

影像處理課本



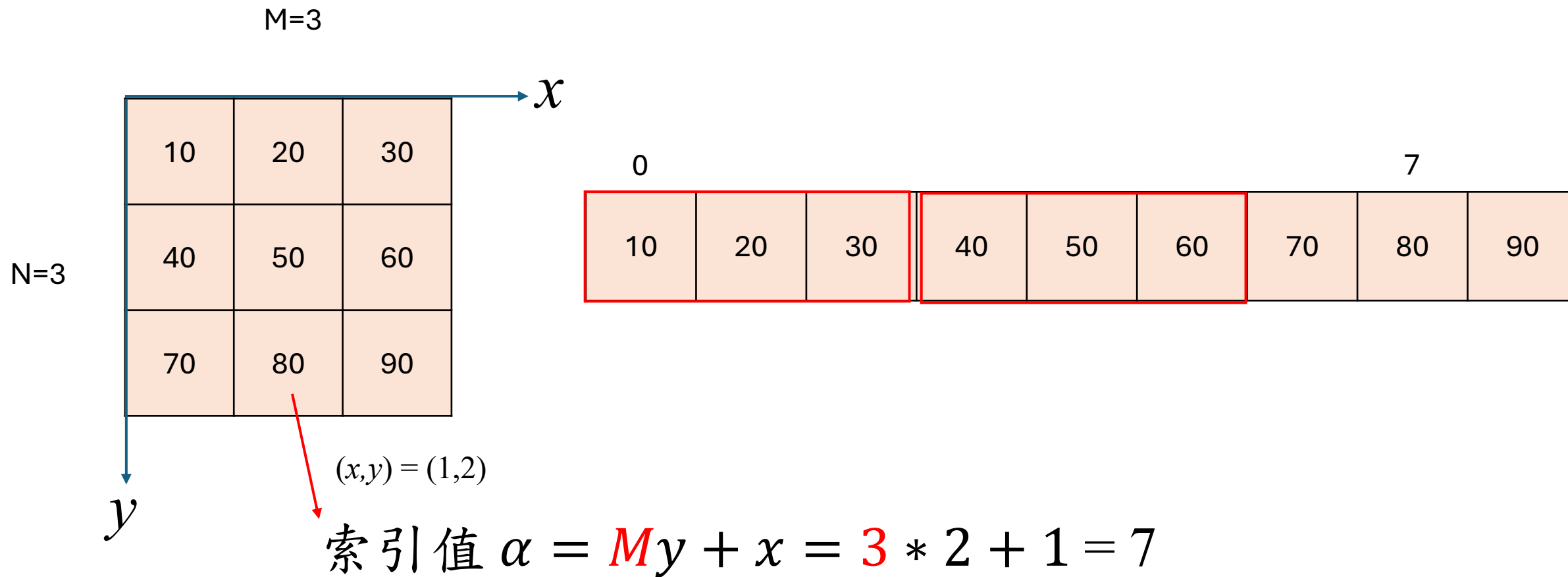
電腦視覺/OpenCV



如果考試的時候有標示不清楚 直接問沒關係

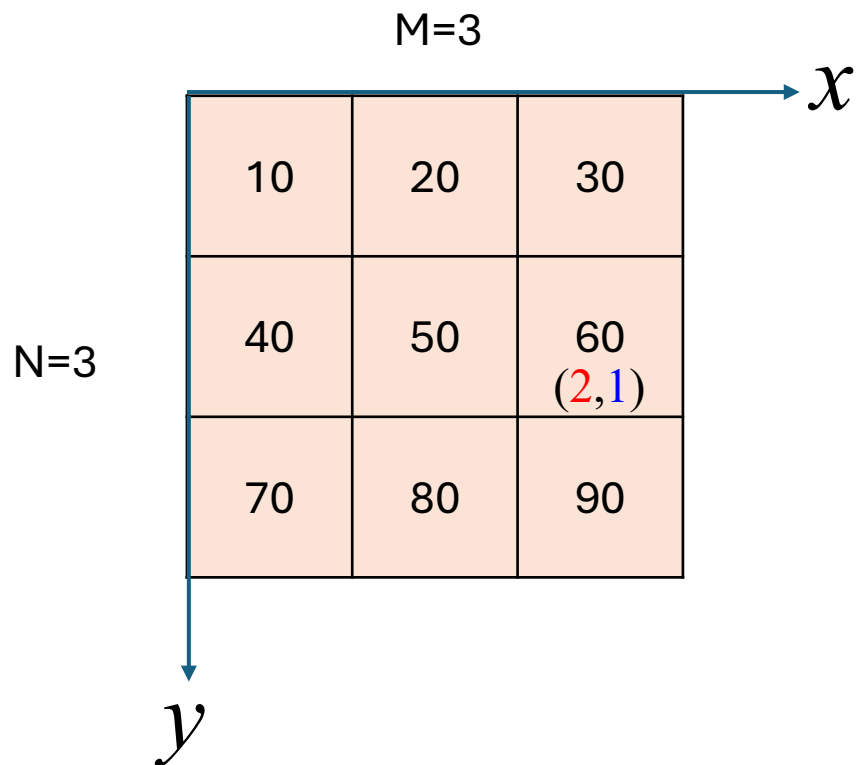


影像索引2D





影像索引2D



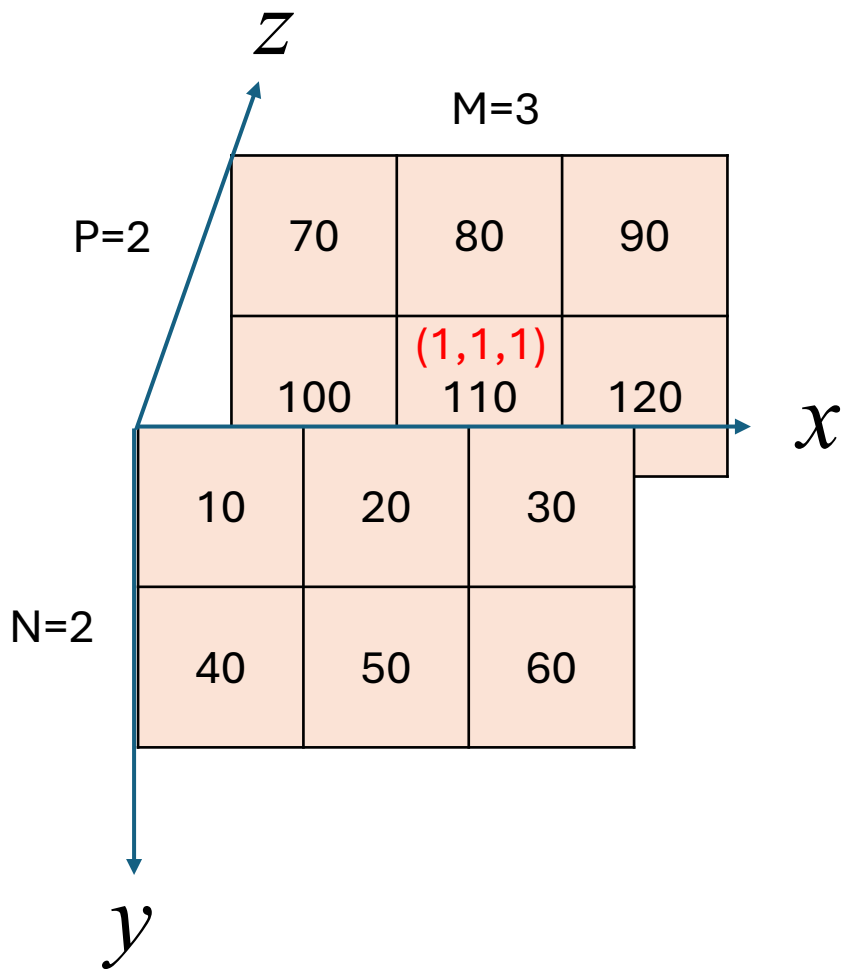
Q: ➤ 已知索引值 α 求 (x,y)
➤ Given the index value α , find (x,y) .

0					$\alpha = 5$			
10	20	30	40	50	60	70	80	90

Sol: ➤ $x = \alpha \bmod M$ $x = 5 \bmod 3 = 2$
➤ $y = \alpha // M$ (取商數) $y = 5 // 3 = 1$



影像索引3D

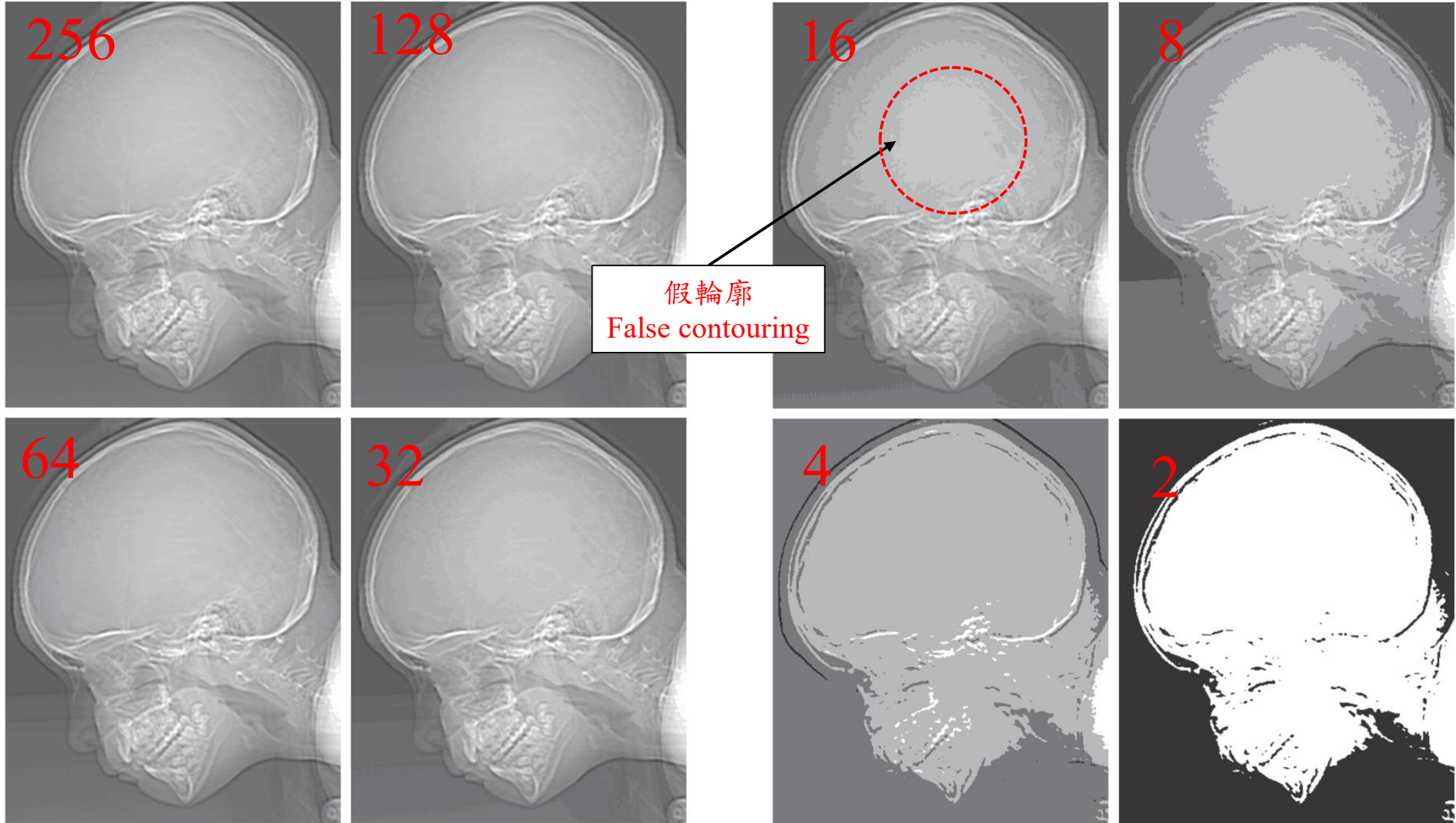


0											10
10	20	30	40	50	60	70	80	90	100	110	120

索引值 $S = \textcolor{red}{x} + My + MNz = x + M(y + Nz)$

- $x = S \bmod MN \bmod M$
- $y = S \bmod MN // M$ (取商數)
- $z = S // MN$ (取商數)

強度解析度



影像內插(Interpolation)

DPI (Dots Per Inch)

930dpi



300dpi



150dpi



72dpi



鄰近差補



雙線性差補



雙立方差補



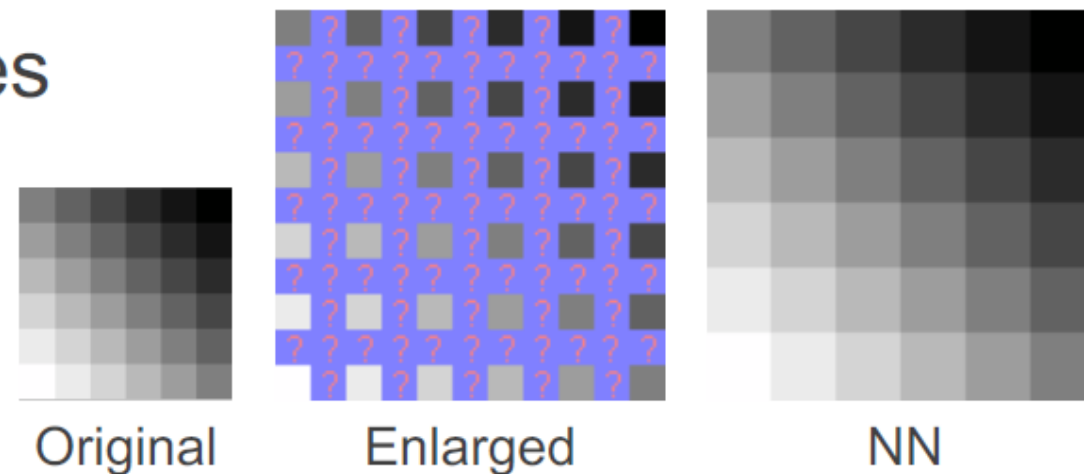
a b c

圖 2.25 (a) 縮小到 72 dpi 再用最近鄰插補放大回其原來 930 dpi 大小的影像。此圖與圖 2.21(d) 相同；(b) 縮小再用雙線性插補放大的影像；(c) 與 (b) 同但用雙立方插補。

影像內插(Interpolation)

- Common interpolation techniques

- Nearest Neighbor (NN)
- Bilinear
- Bicubic

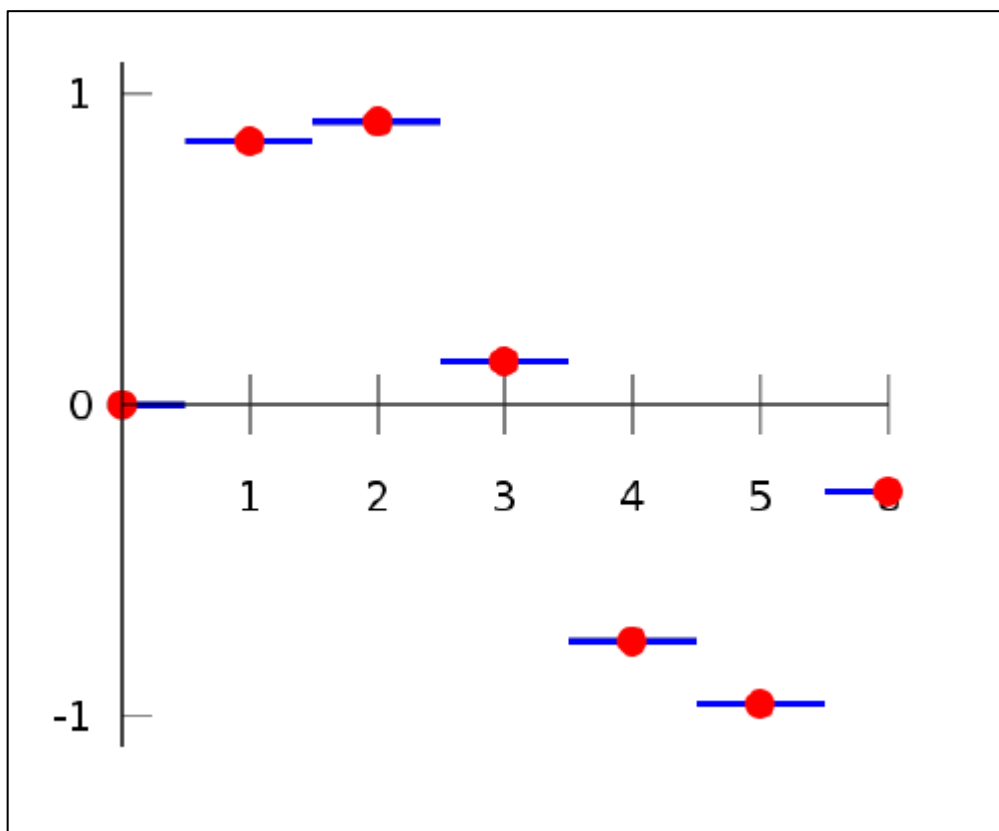


Nearest Neighbor

Bilinear

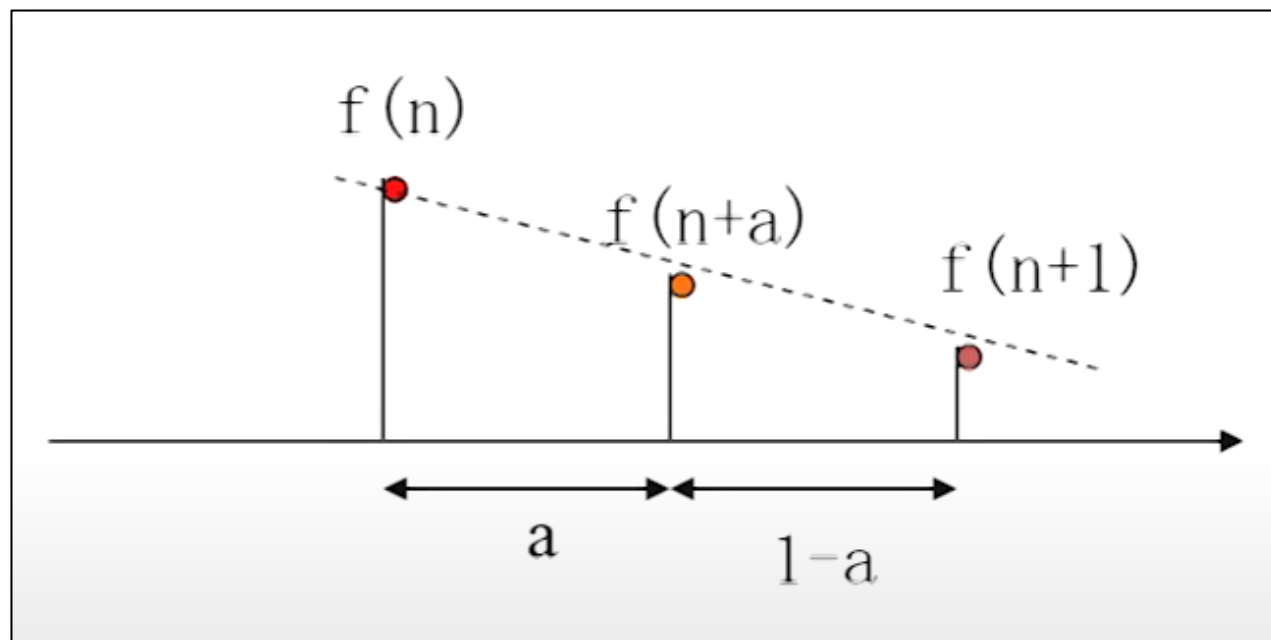
1D-影像內插(Interpolation)

1-D最近鄰內插法



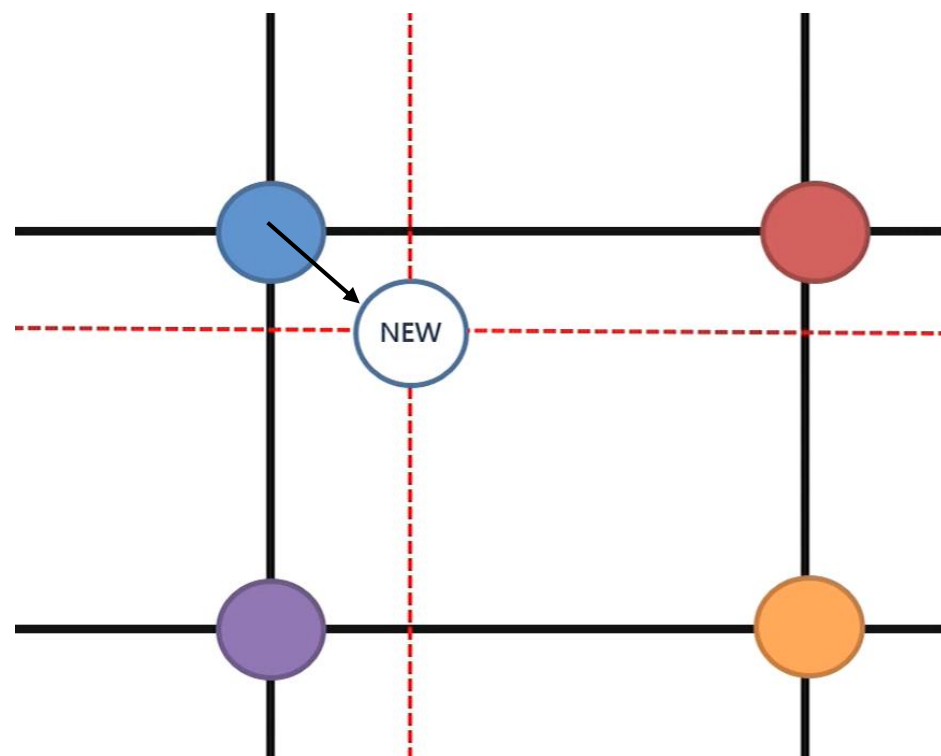
1-D線性內插法

$$f(n+a) = (1-a) \times f(n) + a \times f(n+1)$$



2-D最鄰近內插法 (Nearest Neighbor Interpolation)

2-D最鄰近內插法



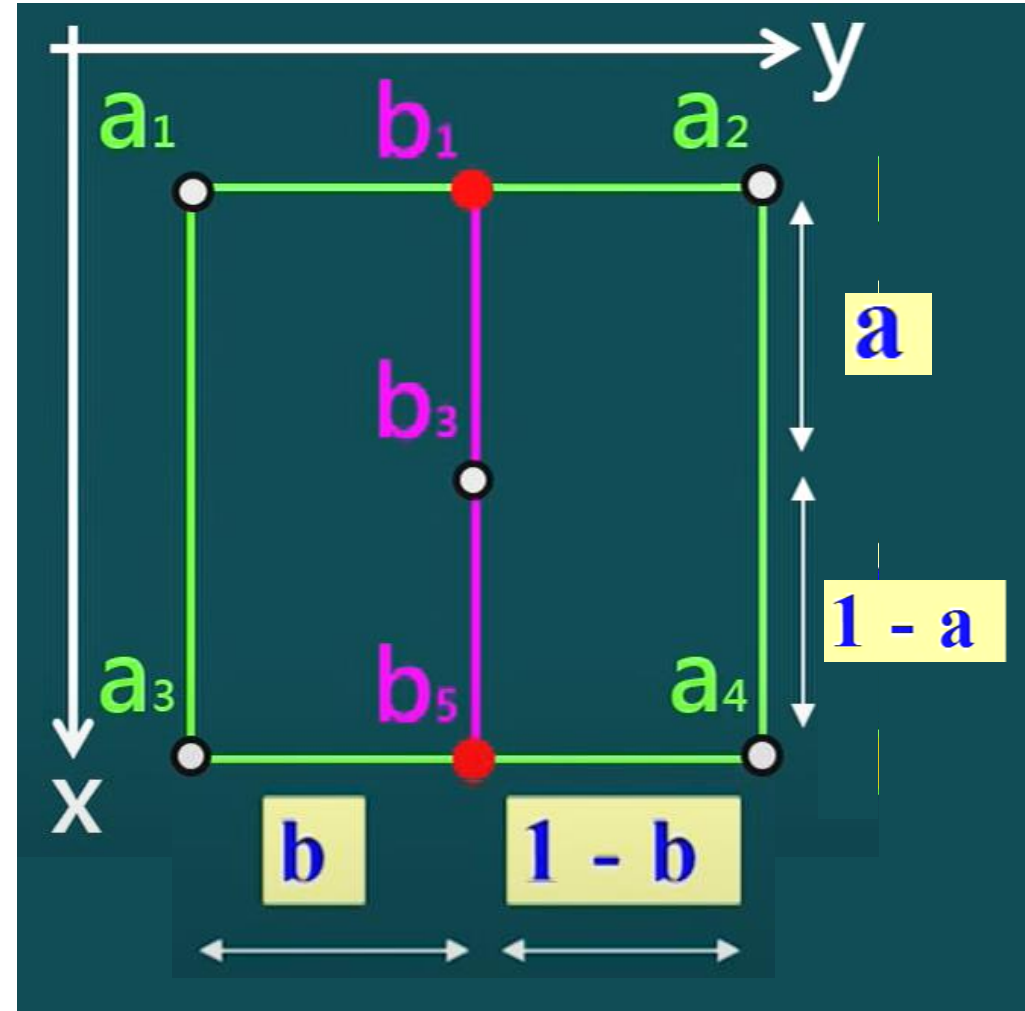
★ 2-D 雙線性內插法(Bilinear Interpolation)

1-D線性內插法

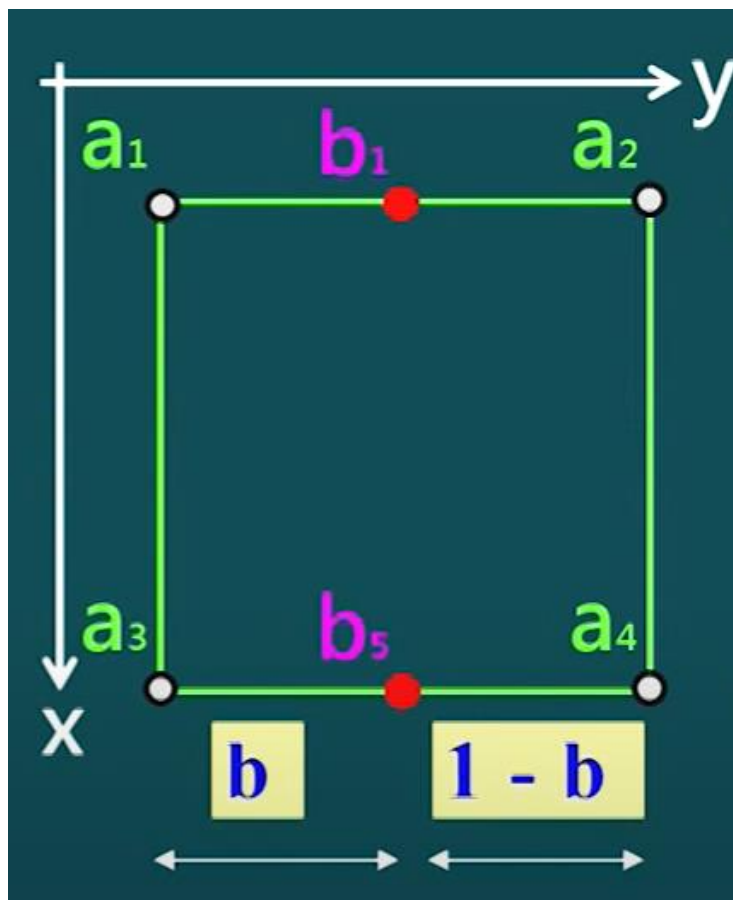
$$f(n + a) = (1 - a) \times f(n) + a \times f(n + 1)$$

2-D線性內插法

$$b_3 = (1 - a) \times [(1 - b) \times a_1 + b \times a_2] \\ + a \times [(1 - b) \times a_3 + b \times a_4]$$



2-D 雙線性內插法 推導(1/3):水平方向內插



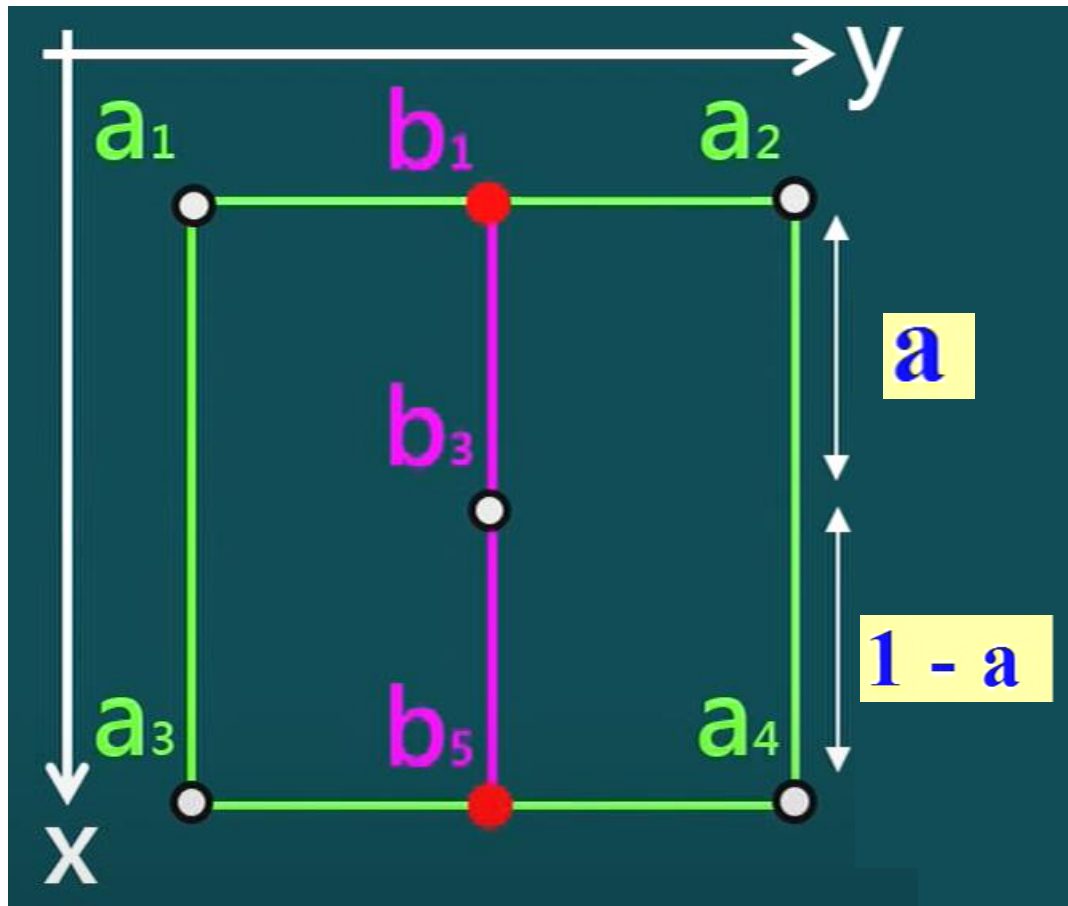
用像素a1、a2 估測出 b1

$$b_1 = (1 - b) \times a_1 + b \times a_2$$

用像素a3、a4 估測出 b5

$$b_5 = (1 - b) \times a_3 + b \times a_4$$

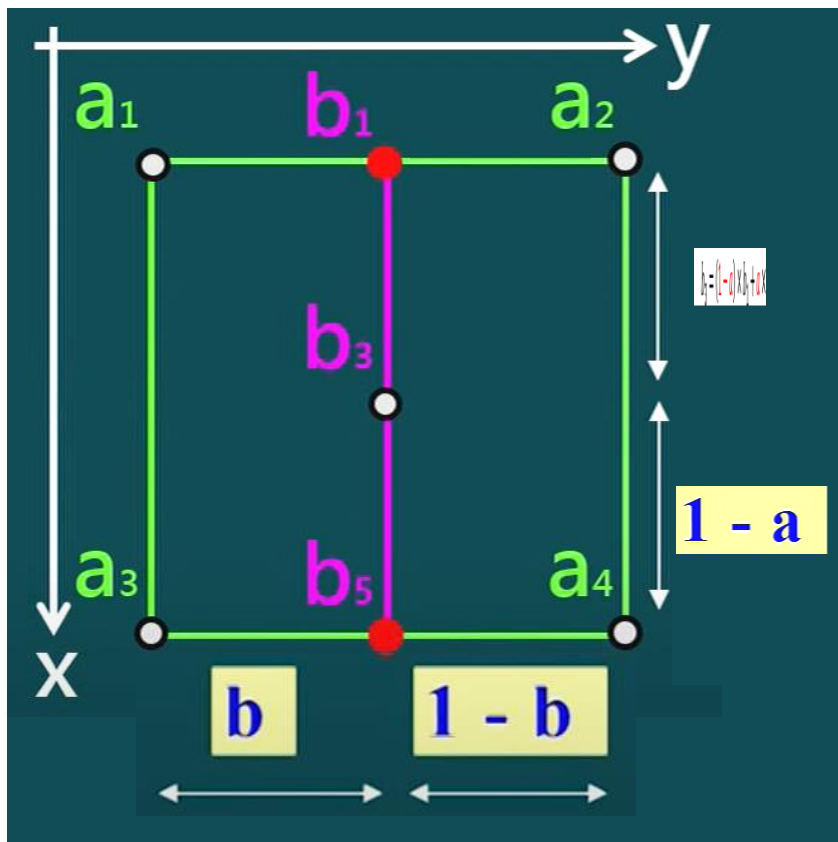
2-D 雙線性內插法 推導(2/3):垂直方向內插



用像素 b_1 、 b_5 估測出 b_3

$$b_3 = (1 - a) \times b_1 + a \times b_5$$

2-D 雙線性內插法 推導(3/3):合併



用像素 a_1 、 a_2 估測出 b_1

$$b_1 = (1 - b) \times a_1 + b \times a_2$$

用像素 a_3 、 a_4 估測出 b_5

$$b_5 = (1 - b) \times a_3 + b \times a_4$$

用像素 b_1 、 b_5 估測出 b_3

$$b_3 = (1 - a) \times b_1 + a \times b_5$$

通式

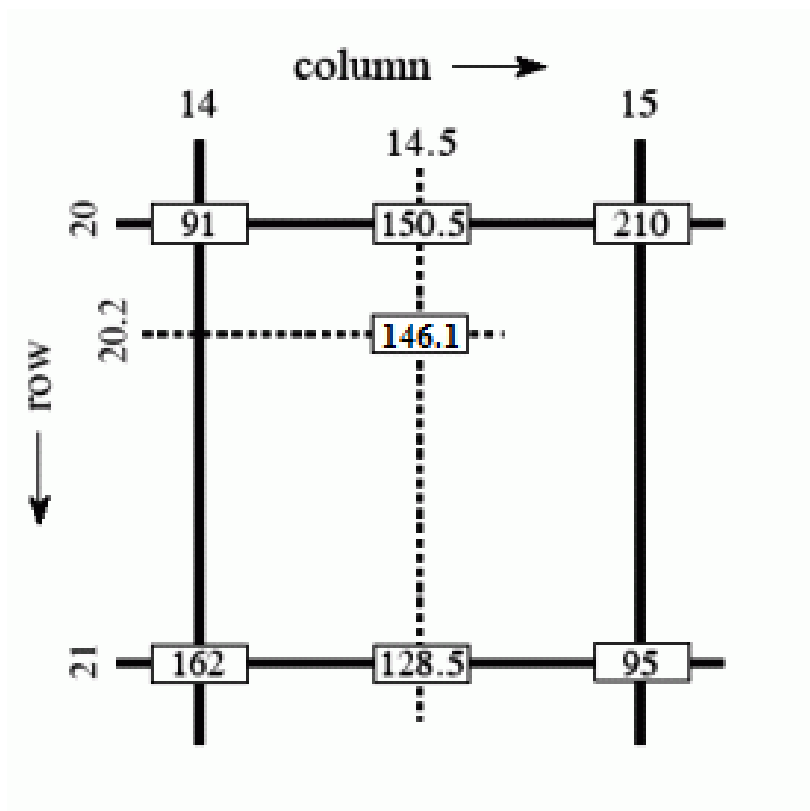
$$b_3 = (1 - a) \times [(1 - b) \times a_1 + b \times a_2] + a \times [(1 - b) \times a_3 + b \times a_4]$$



2-D 雙線性內插法 範例Example 1

https://en.wikipedia.org/wiki/Bilinear_interpolation

Q: Using bilinear interpolation, compute the image intensity at point (20.2, 14.5).



$$I_{20,14.5} = \frac{15 - 14.5}{15 - 14} \cdot 91 + \frac{14.5 - 14}{15 - 14} \cdot 210 = 150.5$$

$$I_{21,14.5} = \frac{15 - 14.5}{15 - 14} \cdot 162 + \frac{14.5 - 14}{15 - 14} \cdot 95 = 128.5$$

$$I_{20.2,14.5} = \frac{21 - 20.2}{21 - 20} \cdot 150.5 + \frac{20.2 - 20}{21 - 20} \cdot 128.5 = 146.1$$



2-D 雙線性內插法 範例Example 2

➤ 40會在哪?

➤ 內插時A點與10跟40的距離?

$\frac{2}{3}$ 、 $\frac{1}{3}$

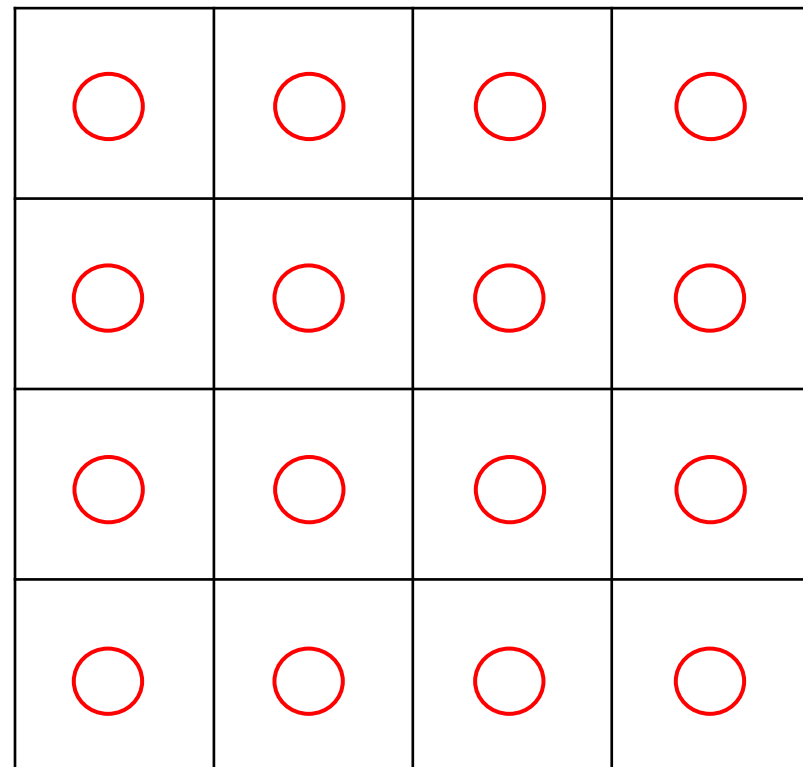
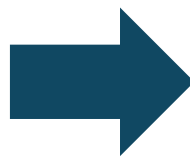
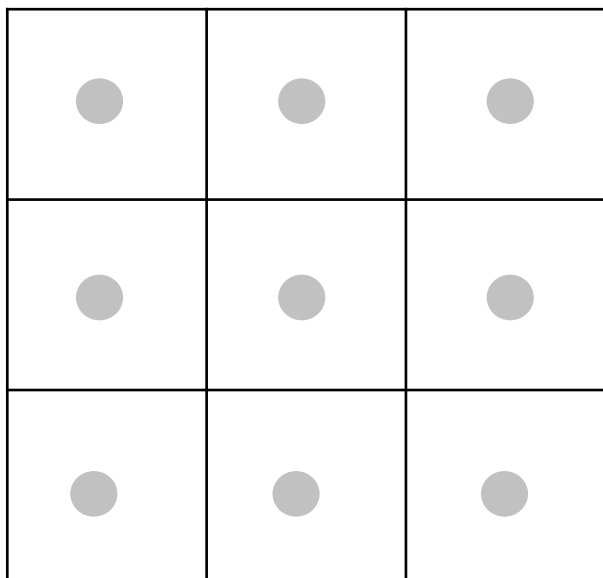
10	20	30
40	50	60
70	80	90



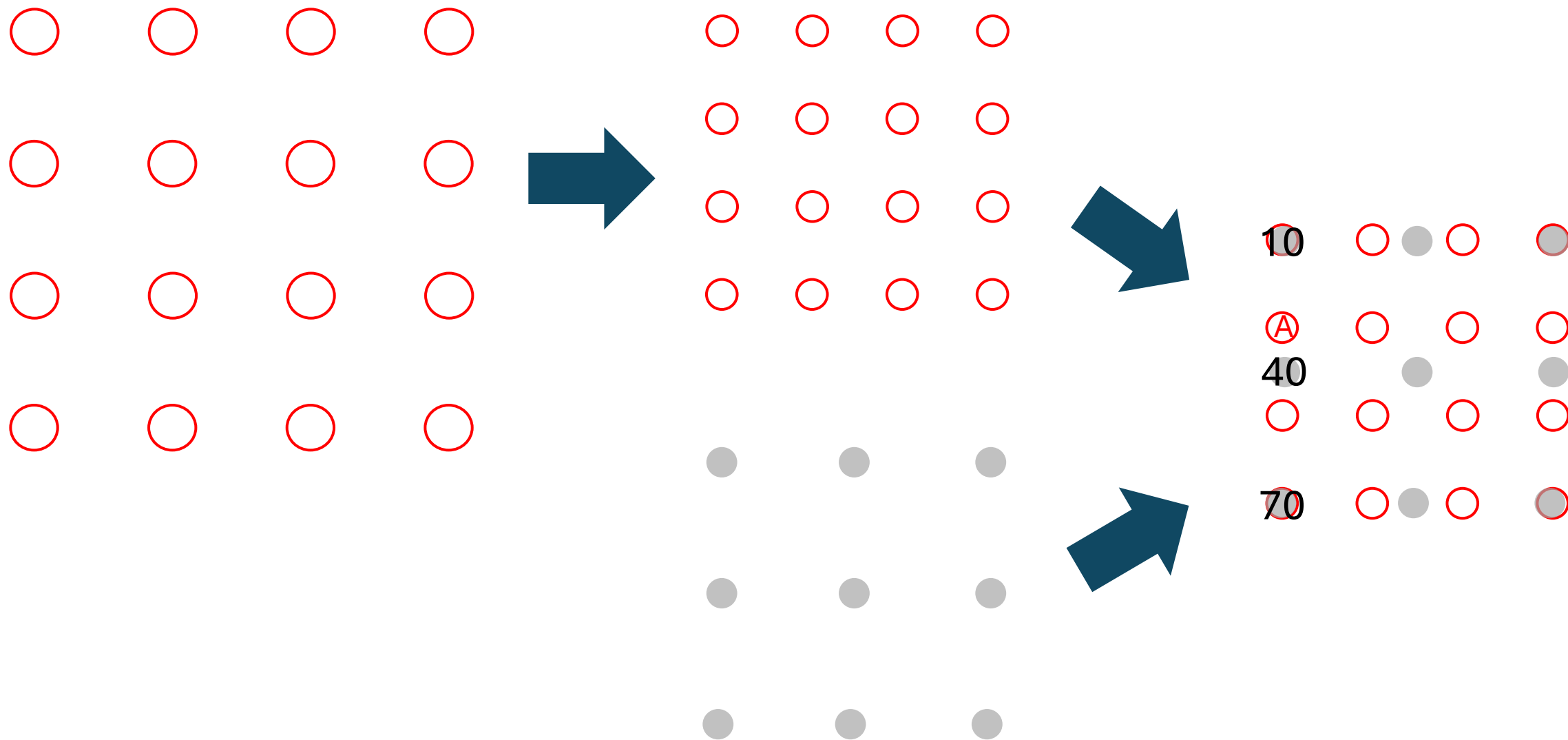
10			30
A			
70			90

2-D 雙線性內插法 範例Example 2

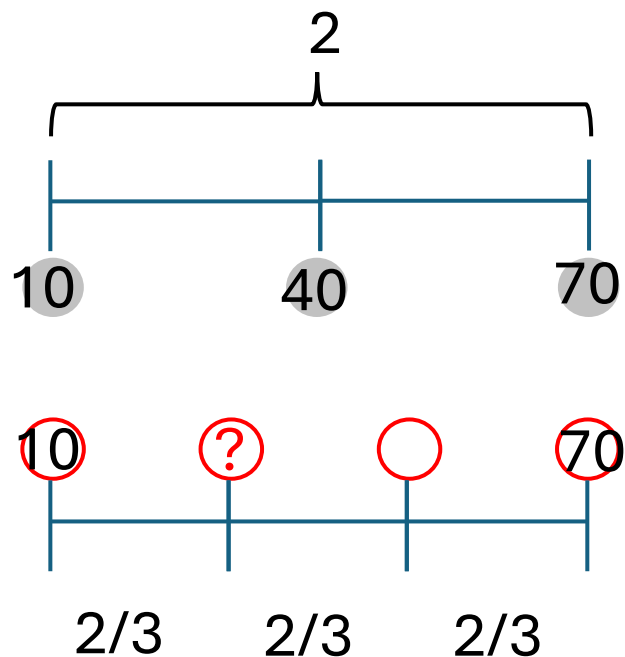
抓像素中心點



四個角落中心點對齊



計算縮放比例



3個像素總長=2

4個像素總長=3

總長度2要平分3個間距

1個間距長度= $\frac{2}{3}$



Step1:計算縮放比例

- 原始圖像尺寸:寬度 $w_1 = 3$, 高度 $h_1 = 3$
- 目標圖像尺寸:寬度 $w_2 = 4$, 高度 $h_2 = 4$

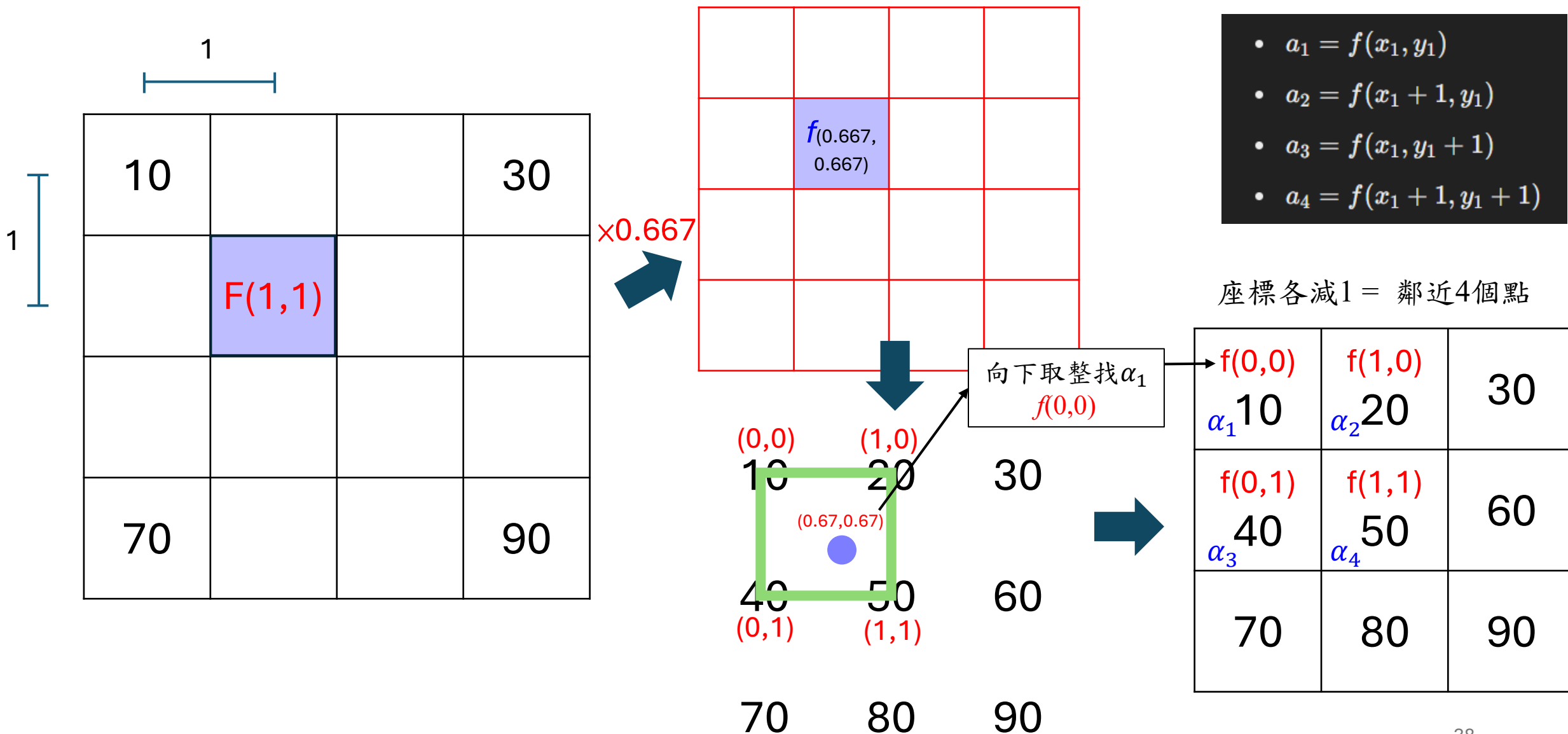
$$S_x = \frac{w_1 - 1}{w_2 - 1} = \frac{3 - 1}{4 - 1} = \frac{2}{3} \approx 0.667$$

$$S_y = \frac{h_1 - 1}{h_2 - 1} = \frac{2}{3} \approx 0.667$$

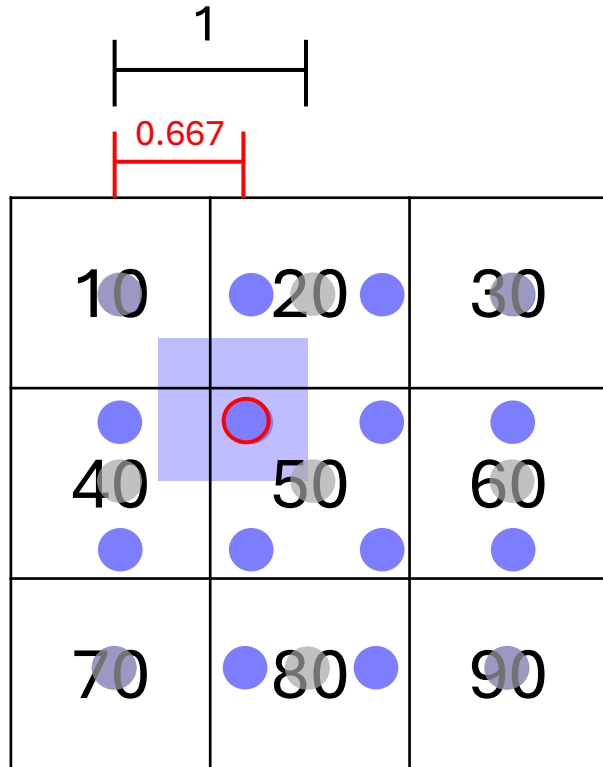
這個就是縮放後
Pixel與Pixel間的距離

F: 目標圖
f: 原圖

Step2: 找鄰近四個點 $\alpha_1 \sim \alpha_4$

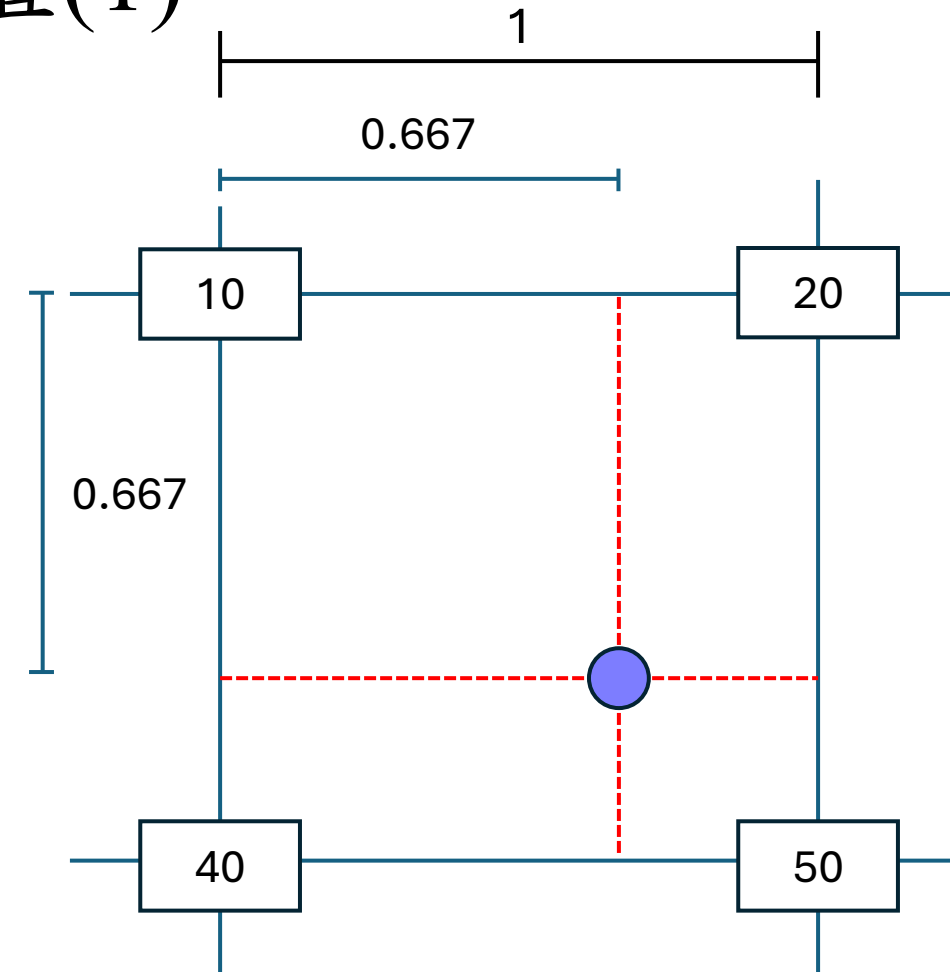
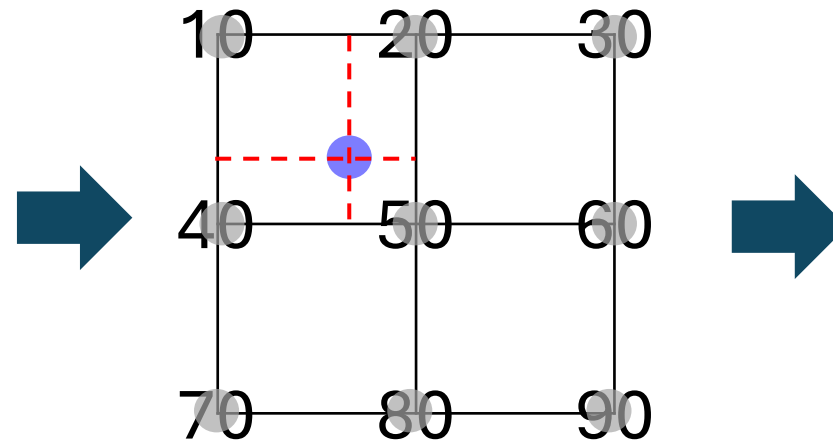


Step3:計算内插値(1)



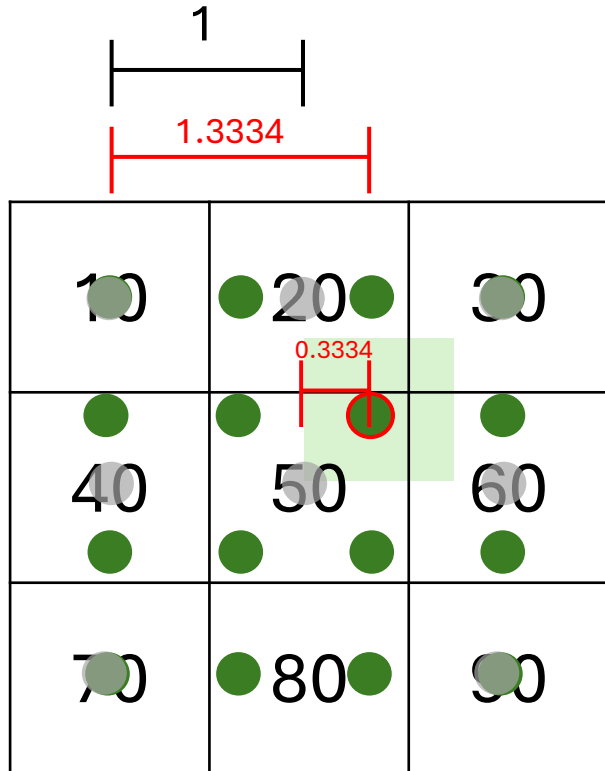
$$0.6667 * 1 = 0.667$$

$$0.6667 * 1 = 0.667$$



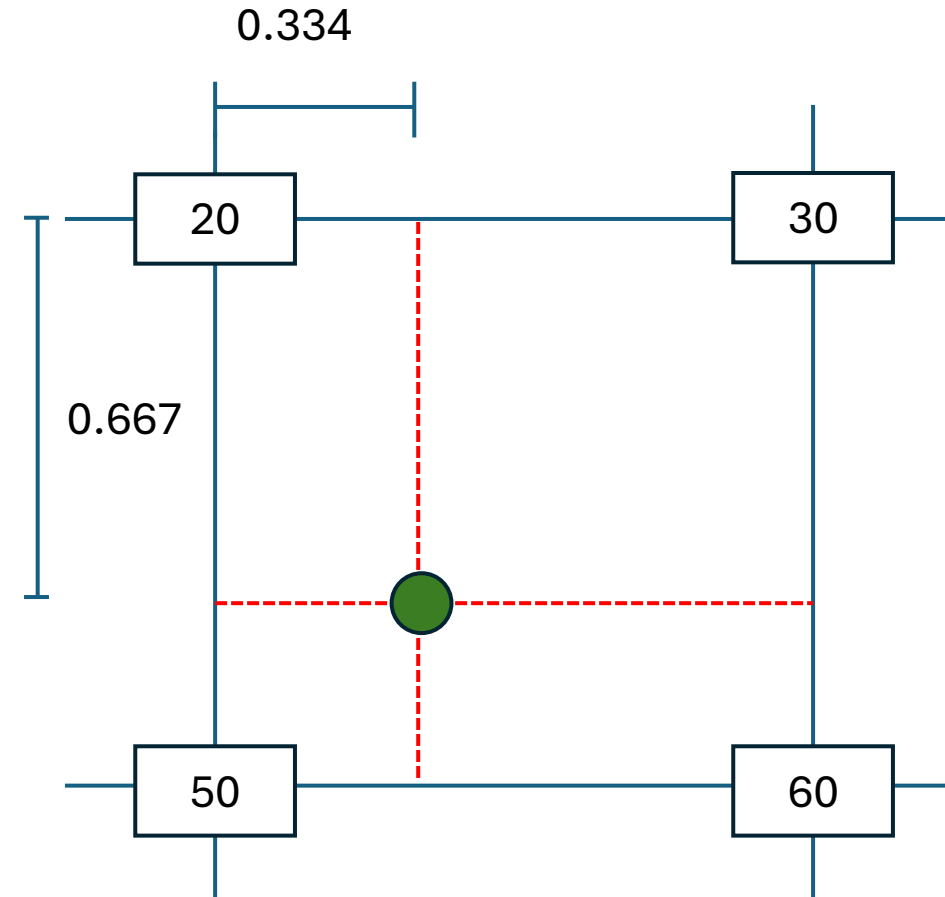
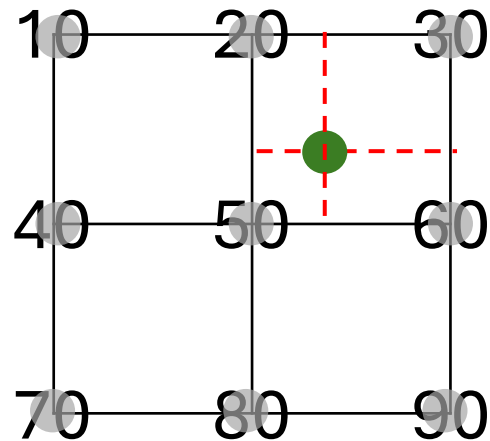
$$F(1,1) = (1-0.6667) \times [(1-0.6667) \times 10 + 0.6667 \times 40] + 0.6667 \times [(1-0.6667) \times 20 + 0.6667 \times 50] = 36.6672$$

Step3:計算內插值(2)



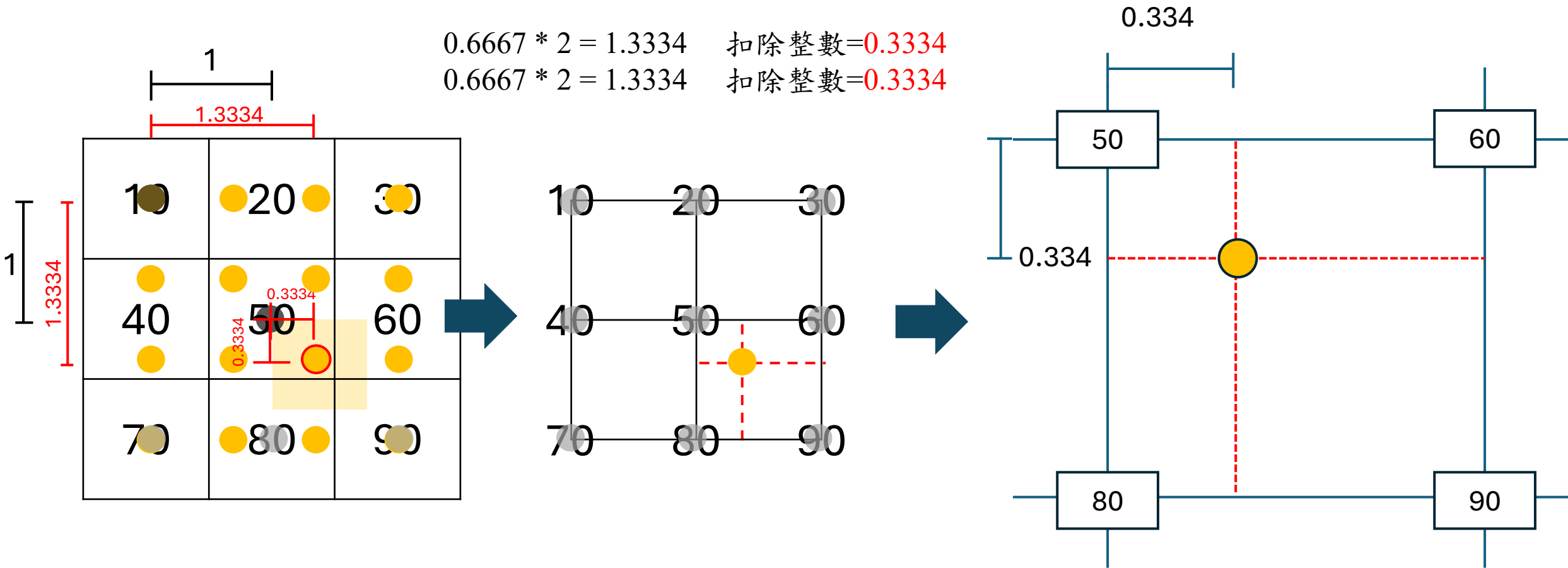
$$0.6667 * 2 = 1.3334 \quad \text{扣除整數} = 0.3334$$

$$0.6667 * 1 = 0.667$$



$$F(2,1) = (1-0.3334) \times [(1-0.6667) \times 20 + 0.6667 \times 50] + 0.3334 \times [(1-0.6667) \times 30 + 0.6667 \times 60] = 43.3352$$

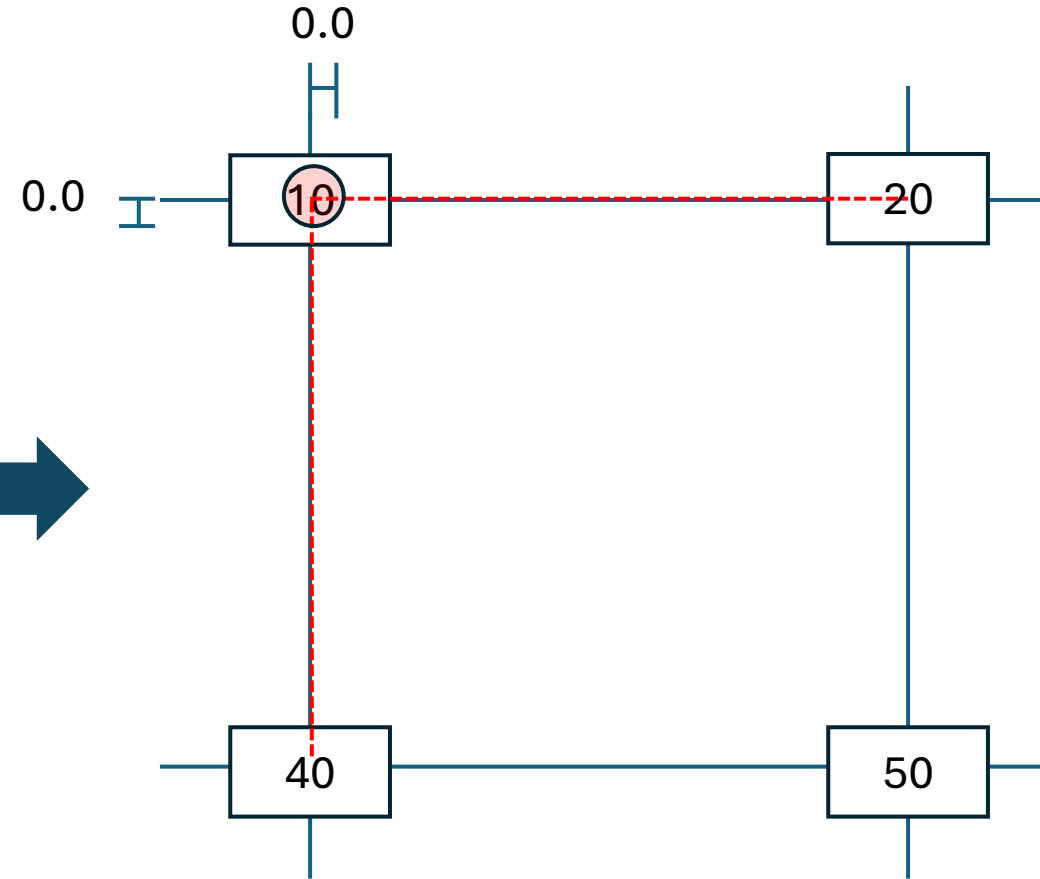
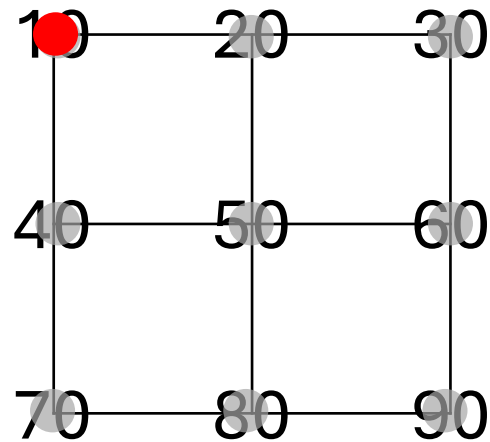
Step3:計算內插值(3)



Step3:計算内插値(4)

$$0.6667 \times 0 = 0$$
$$0.6667 \times 0 = 0$$

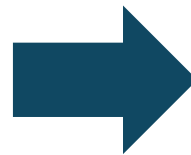
10	20	30
40	50	60
70	80	90



$$F(0,0) = 1 \times [0 \times 40 + 1 \times 10] +$$
$$0 \times [(0 \times 50 + 1 \times 20)] = 10$$

完成!

10	20	30
40	50	60
70	80	90



10	16.6	23.3	30
30	36.6	43.3	50
50	56.6	63.3	70
70	76.6	83.3	90

2-D 雙線性內插法 整理

Step1:縮放比例 (Scaling Ratio)

- 原始圖像尺寸:寬度 w_1 , 高度 h_1
- 目標圖像尺寸:寬度 w_2 , 高度 h_2

$$S_x = \frac{w_1 - 1}{w_2 - 1} \quad S_y = \frac{h_1 - 1}{h_2 - 1}$$

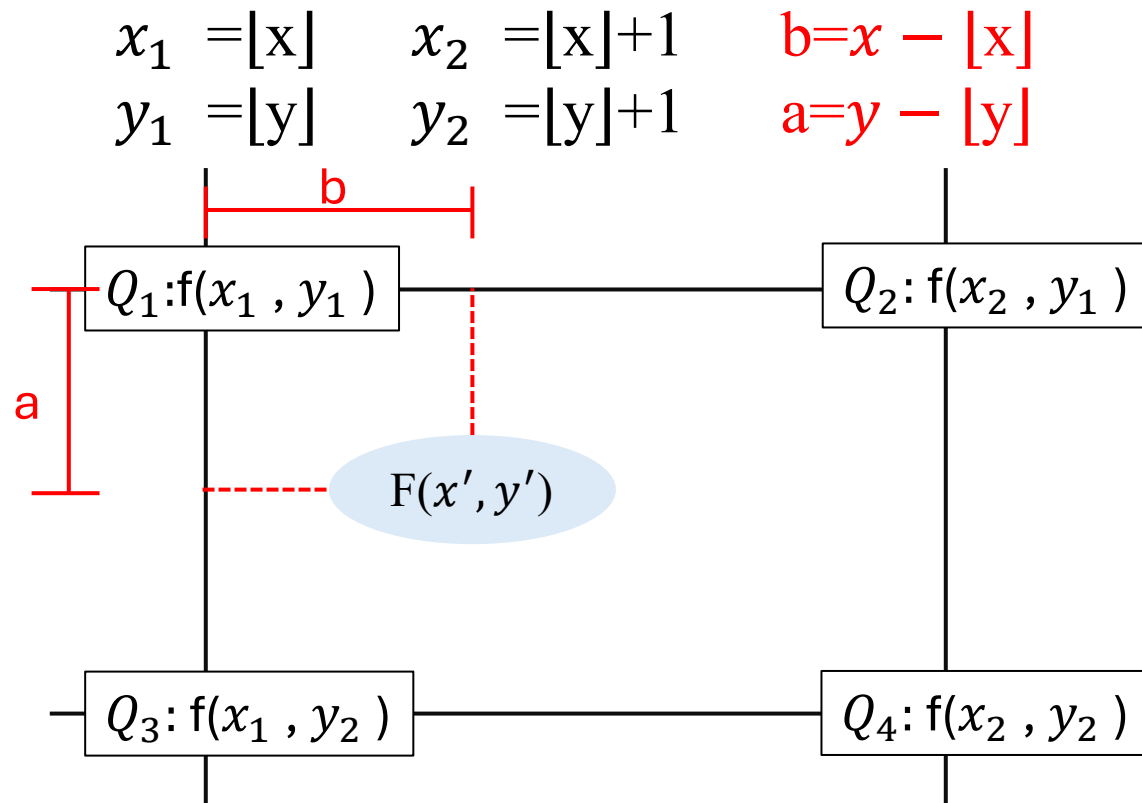
Step2:映射到原圖

要算目標圖 $F(x', y')$ 的值
先算映射到原圖的 $f(x, y)$

$$x = x' \times S_x$$

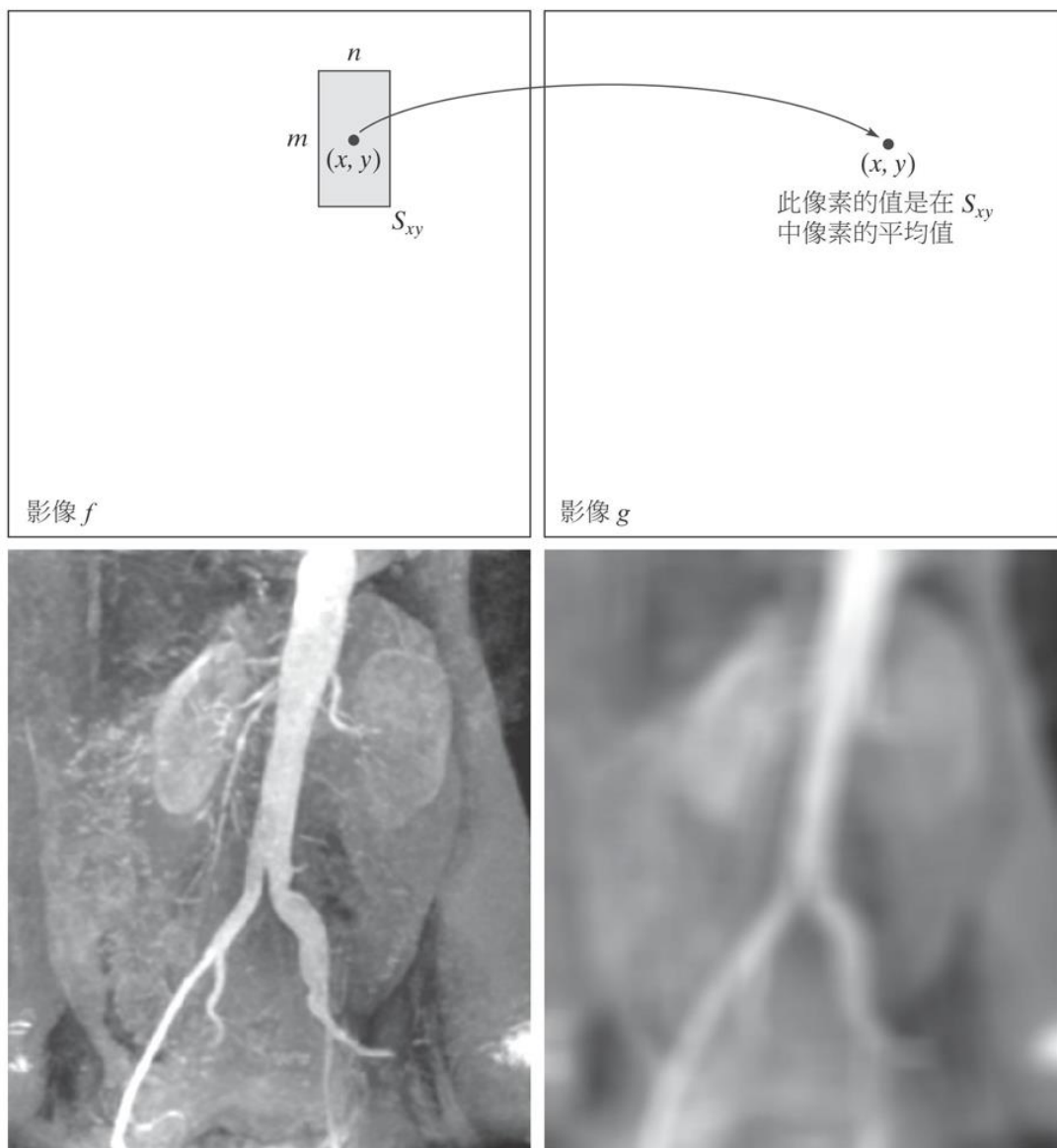
$$y = y' \times S_y$$

Step3:計算目標圖值



$$F(x', y') = (1-b) \times [(1-a) \times Q_1 + a \times Q_3] + b \times [(1-a) \times Q_2 + a \times Q_4]$$

2.5 空間運算：鄰域運算



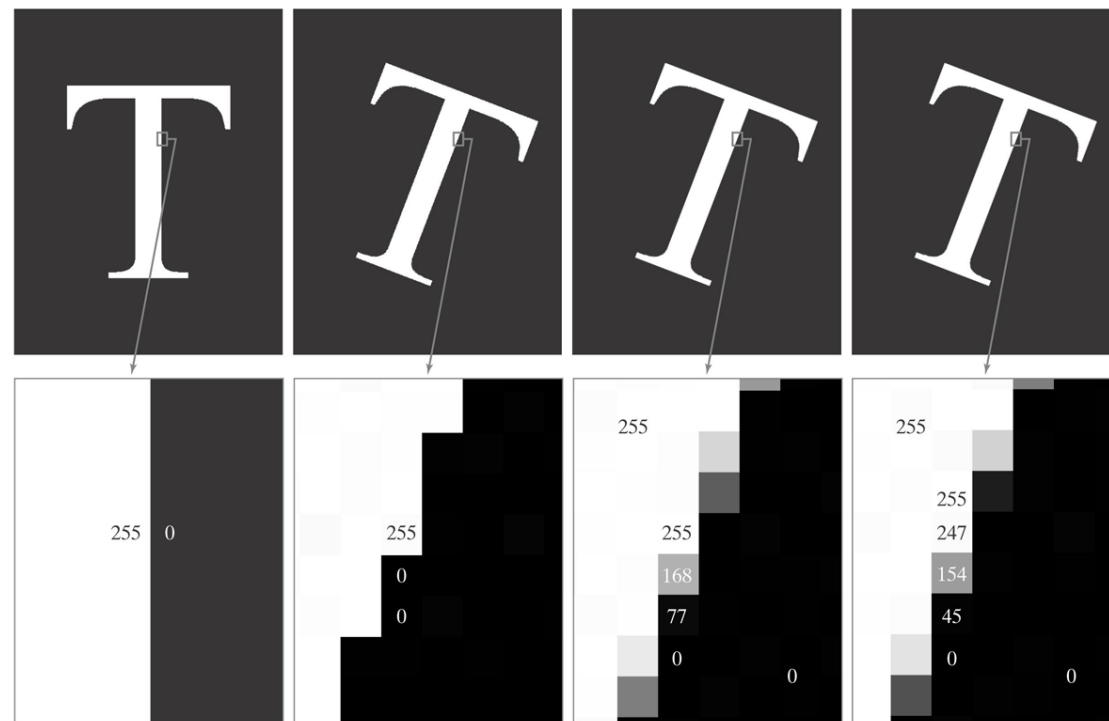
平均

$$g(x, y) = \frac{1}{mn} \sum_{(r, c) \in S_{xy}} f(r, c)$$

2.5 空間運算：仿射轉換

表 2.1 依據 (2-23) 式的仿射轉換

轉換名稱	仿射矩陣 A	座標方程式	例 子
恆等式	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = y$	
尺度調整 / 翻轉 (對於翻轉，將一個尺度因子設成 -1 且其它設成 0)	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = c_x x$ $y' = c_y y$	
旋轉 (對原點)	$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x \cos\theta - y \sin\theta$ $y' = x \sin\theta + y \cos\theta$	
平移	$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + t_x$ $y' = y + t_y$	
切變 (垂直)	$\begin{bmatrix} 1 & s_v & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + s_v y$ $y' = y$	
切變 (水平)	$\begin{bmatrix} 1 & 0 & 0 \\ s_h & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = s_h x + y$	



a b c d
e f g h

圖 2.29 (a) 字母 T 的 541×421 影像；(b) 用最近鄰插補法做強度指定給經旋轉 -21° 的影像；(c) 用雙線性插補法使影像旋轉 -21° ；(d) 用雙立方插補法使影像旋轉 -21° 。(e)~(h) 放大部分 (每個方形是一個像素，而所顯示的數字是強度值)。

$$[x \ y \ 1] = [v \ w \ 1] \mathbf{T} = [v \ w \ 1] \begin{bmatrix} t_{11} & t_{12} & 0 \\ t_{21} & t_{22} & 0 \\ t_{31} & t_{32} & 1 \end{bmatrix}$$

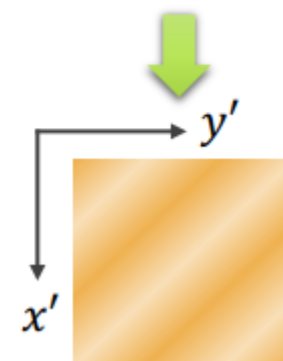
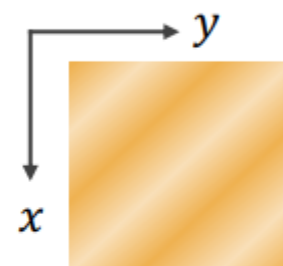
(1)恒等式

- Identity transformation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Identity

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



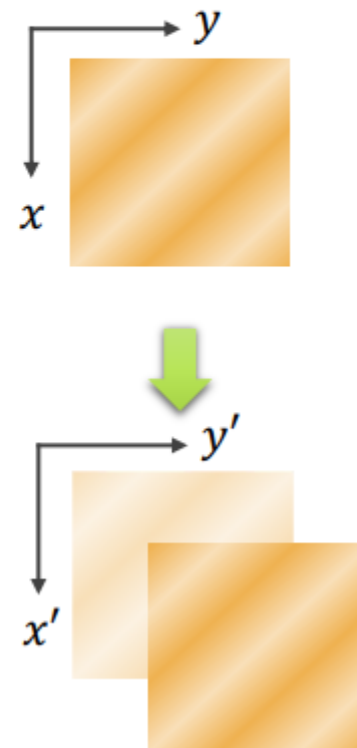
(2) 平移

- Affine transformation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Translation

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$



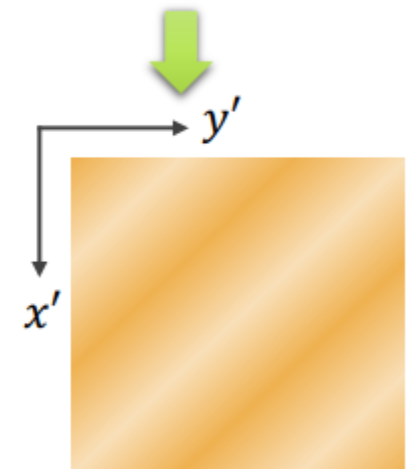
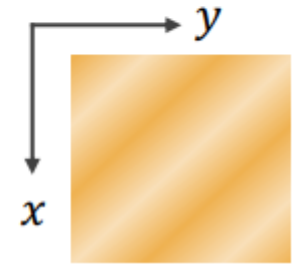
(3) 放大縮小

- Affine transformation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Scaling

$$\mathbf{A} = \begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

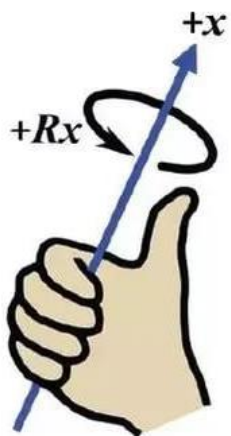


(4) 旋轉

- Affine transformation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

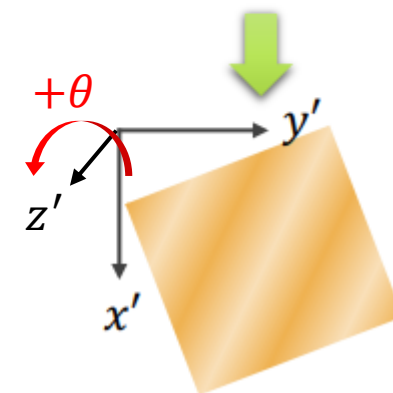
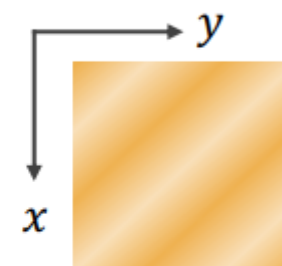
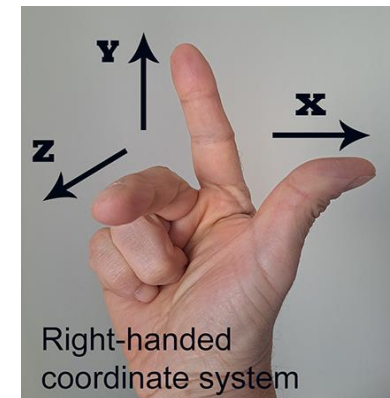
- Rotation



大拇指方向 = 旋轉軸正方向
四指彎曲方向 = 正旋轉方向

$$\mathbf{A} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

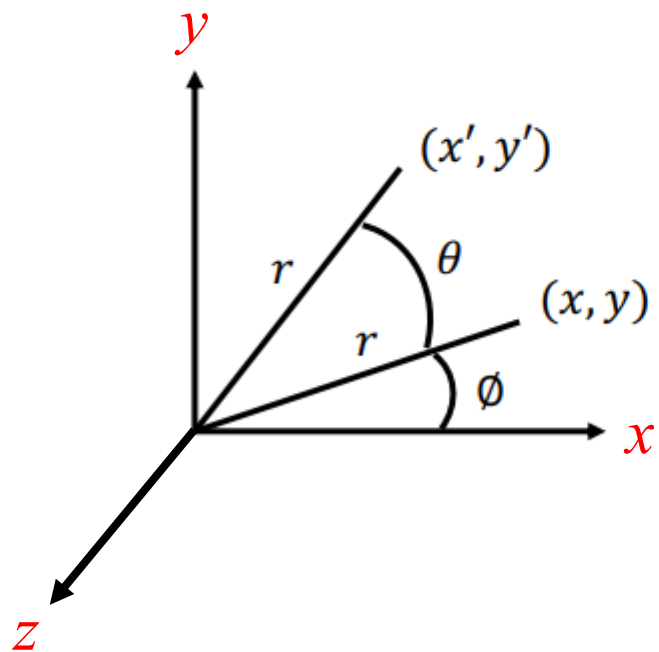
找出x,y之後就能定義z方向



旋轉-推導過程

- Rotation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



$$x = r \cos \phi$$

$$y = r \sin \phi$$

$$x' = r \cos(\phi + \theta)$$

$$y' = r \sin(\phi + \theta)$$

代入

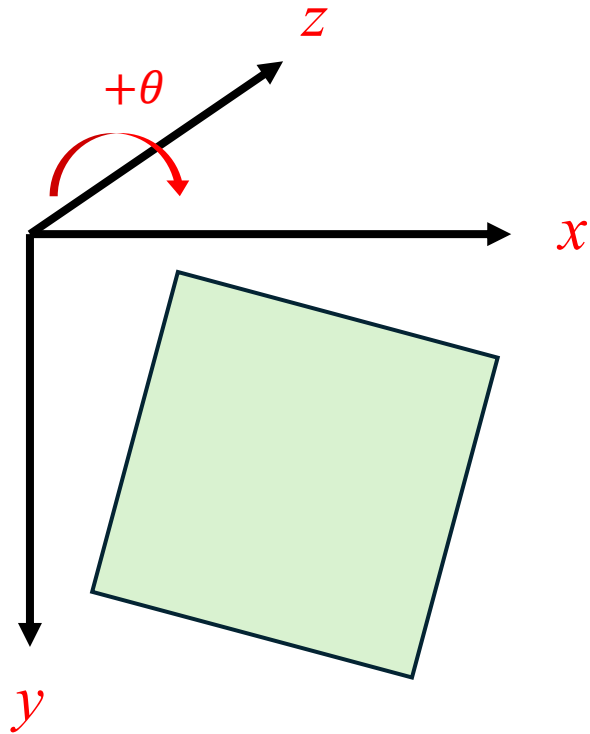
$$x' = r \cos(\phi) \cos(\theta) - r \sin(\phi) \sin(\theta)$$

$$y' = r \sin(\phi) \cos(\theta) + r \cos(\phi) \sin(\theta)$$

$$x' = x \cos(\theta) - y \sin(\theta)$$

$$y' = x \sin(\theta) + y \cos(\theta)$$

旋轉-電腦視覺座標



如果是電腦視覺用的座標，z軸方向通常向後為正，旋轉軸為順時針為正

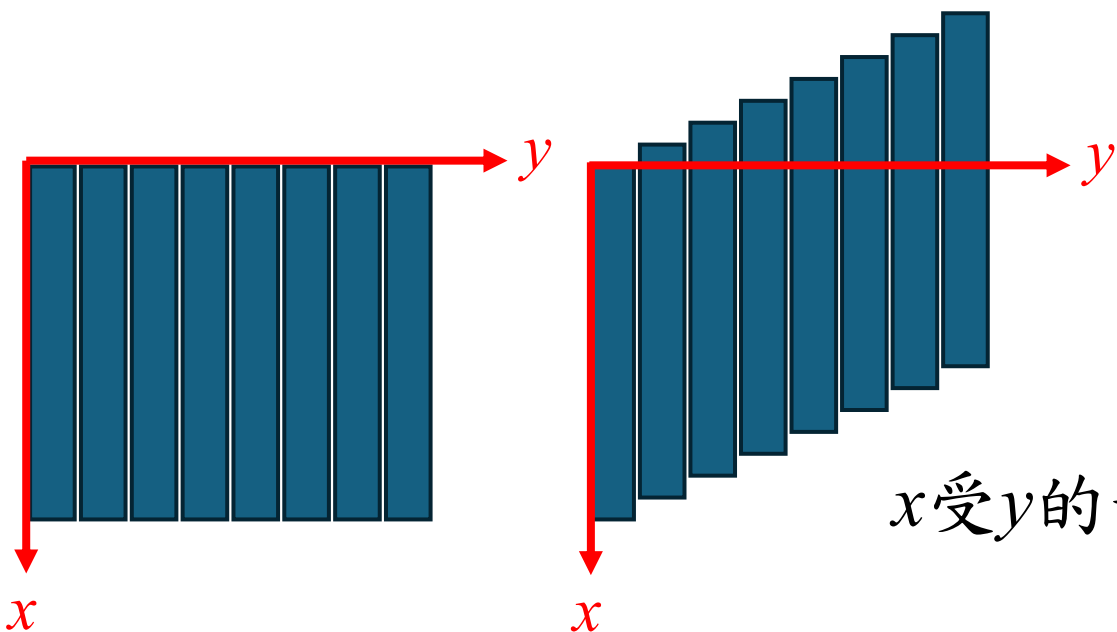
In computer vision, the z-axis is usually positive backward, with clockwise rotation taken as positive.

(5) 切變(垂直)

- Affine transformation

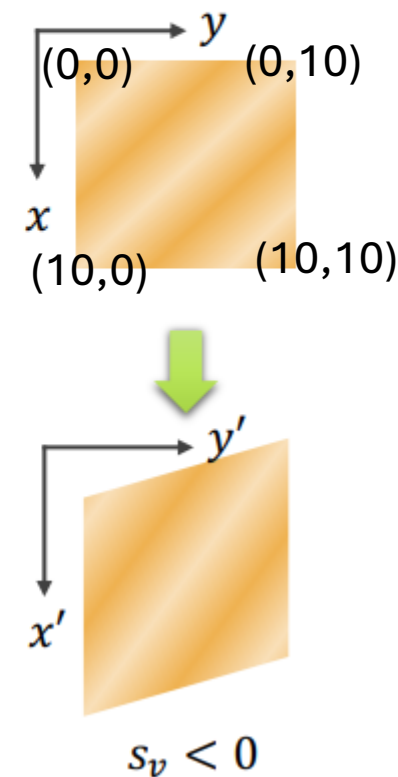
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Shearing (vertical)



$$\mathbf{A} = \begin{bmatrix} 1 & s_v & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

x 受 y 的影響 越遠影響越大



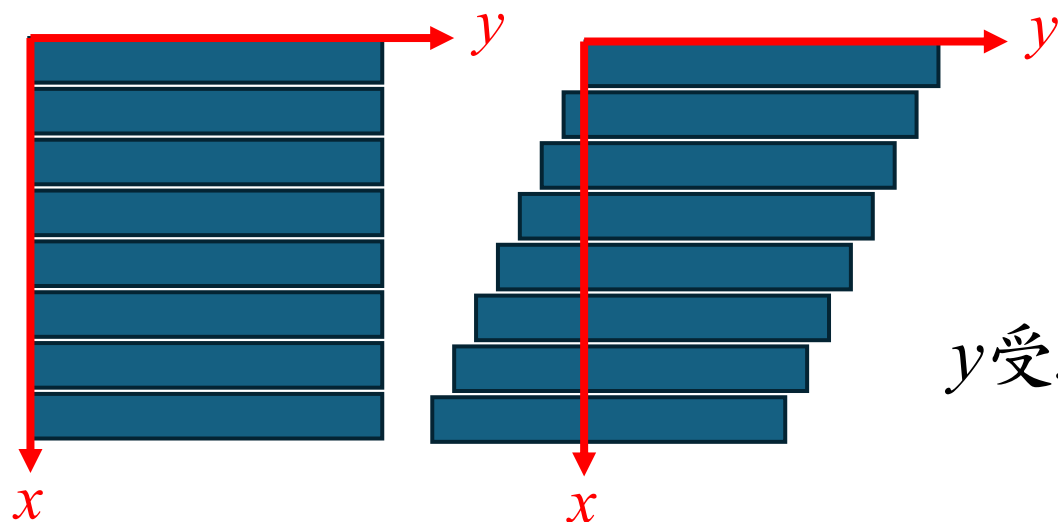
(5) 切變(水平)

- Affine transformation

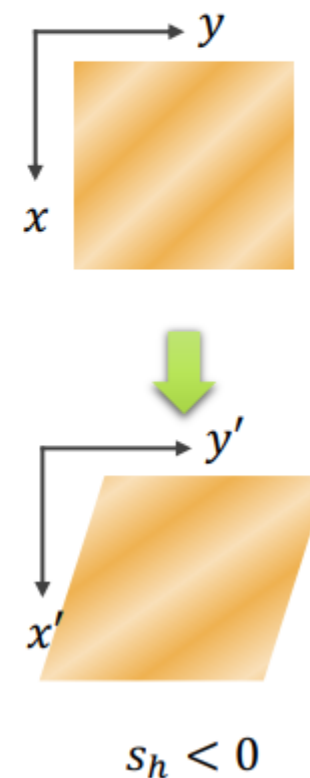
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Shearing (horizontal)

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ s_h & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



y 受 x 的影響 越遠影響越大



影像還原

- Forward mapping

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Backward mapping

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \mathbf{A}^{-1} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

3x3 反矩陣 推導

第一種:直覺

平移&反平移

$$T = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

$$T^{-1} = \begin{bmatrix} 1 & 0 & -t_x \\ 0 & 1 & -t_y \\ 0 & 0 & 1 \end{bmatrix}$$

縮放&反縮放

$$S = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$S^{-1} = \begin{bmatrix} \frac{1}{s_x} & 0 & 0 \\ 0 & \frac{1}{s_y} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

旋轉&反旋轉

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R(\theta)^{-1} = R(-\theta) = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3x3 反矩陣 推導

第二種：餘因子公式

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

$$T = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

$$\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32})$$

$$- a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

- $C_{11} = + \det \begin{bmatrix} 1 & t_y \\ 0 & 1 \end{bmatrix} = 1$
- $C_{12} = - \det \begin{bmatrix} 0 & t_y \\ 0 & 1 \end{bmatrix} = -(0 - 0) = 0$
- $C_{13} = + \det \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = 0$
- $C_{21} = - \det \begin{bmatrix} 0 & t_x \\ 0 & 1 \end{bmatrix} = -(0 - 0) = 0$
- $C_{22} = + \det \begin{bmatrix} 1 & t_x \\ 0 & 1 \end{bmatrix} = 1$
- $C_{23} = - \det \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = -(0 - 0) = 0$
- $C_{31} = + \det \begin{bmatrix} 0 & t_x \\ 1 & t_y \end{bmatrix} = 0 \cdot t_y - t_x \cdot 1 = -t_x$
- $C_{32} = - \det \begin{bmatrix} 1 & t_x \\ 0 & t_y \end{bmatrix} = -(1 \cdot t_y - t_x \cdot 0) = -t_y$
- $C_{33} = + \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1$

$$\text{adj}(T) = C^T = \begin{bmatrix} 1 & 0 & -t_x \\ 0 & 1 & -t_y \\ 0 & 0 & 1 \end{bmatrix}.$$

$$T^{-1} = \frac{1}{\det(T)} \text{adj}(T) = \begin{bmatrix} 1 & 0 & -t_x \\ 0 & 1 & -t_y \\ 0 & 0 & 1 \end{bmatrix}.$$

仿射矩陣-正向

假設我有一個旋轉矩陣 $R = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$

一個平移矩陣 $T = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$

一個影像某個點 $A = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$

因為先平移=(TA)

再(TA)旋轉 = R(TA)

所以整合起來會是(RT)A

(越先算的會越靠近 $\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$)

如果是先平移再旋轉，不是



$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

先平移再旋轉，是



$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

EX:仿射矩陣-反向 轉回去

- 假設一個影像先R順時針正旋轉 $\theta=45$ ，再T平移 ($t_x=3, t_y=2$)。
- 請寫出影像的仿射矩陣以及將影像變回來的逆向仿射矩陣。

正向仿射矩陣：

$$\begin{array}{c} \text{再平移T} \quad \text{先旋轉R} \\ \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 3 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \end{array}$$

$$\begin{array}{c} \text{再平移} \quad \text{先旋轉} \\ \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

逆向仿射矩陣：

$$\begin{array}{c} \text{再轉回來} \quad \text{先平移回來} \\ \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & -\frac{5\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

$$\begin{array}{c} \text{再反旋轉} \quad \text{先反平移} \\ \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -t_x \\ 0 & 1 & -t_y \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

小考會考

- 線性索引
- 雙線性內插
- 仿射轉換